repABC Origin Maps — SECONDARY

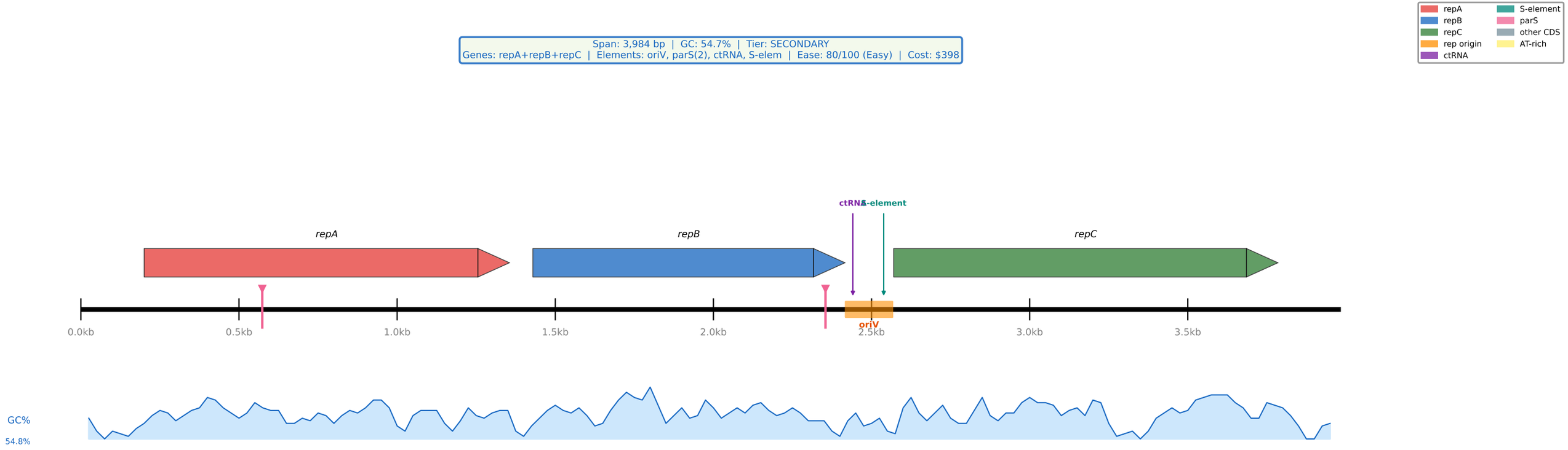
86 origins | 351,555 bp total

Synthesis ease: {'Easy': np.int64(80), 'Moderate': np.int64(6)} | Est. cost: $35,071

Rep classes: 21 | Strains: 28

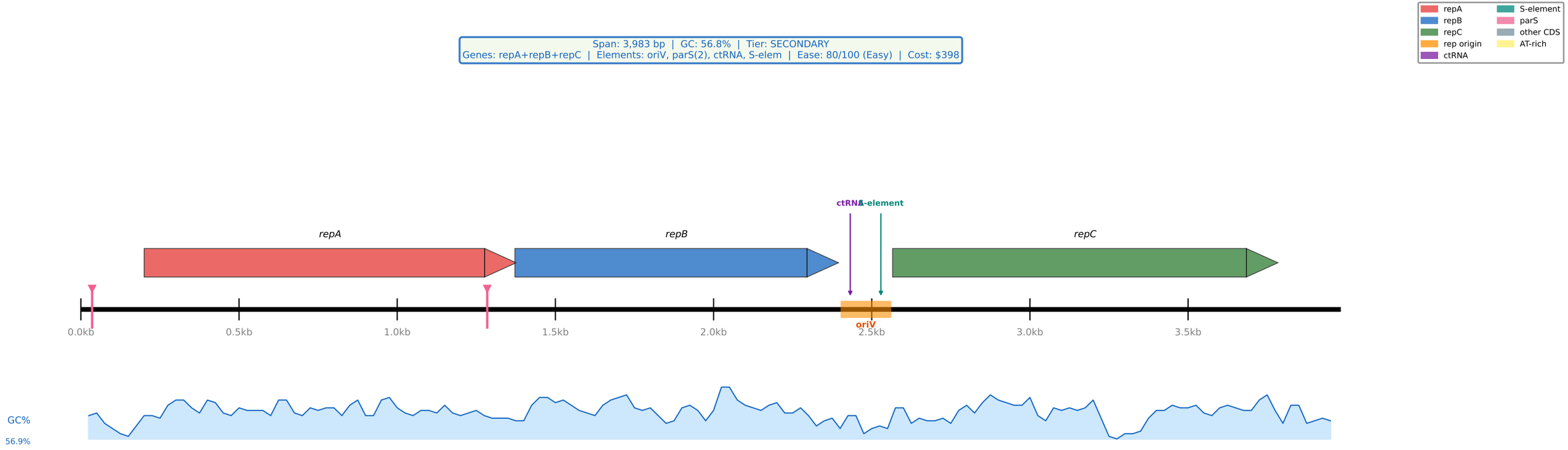
AMS008_plasmid2 | unknown | Agrobacterium tumefaciens

Span: 3,984 bp | GC: 54.7% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: oriV, parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $398



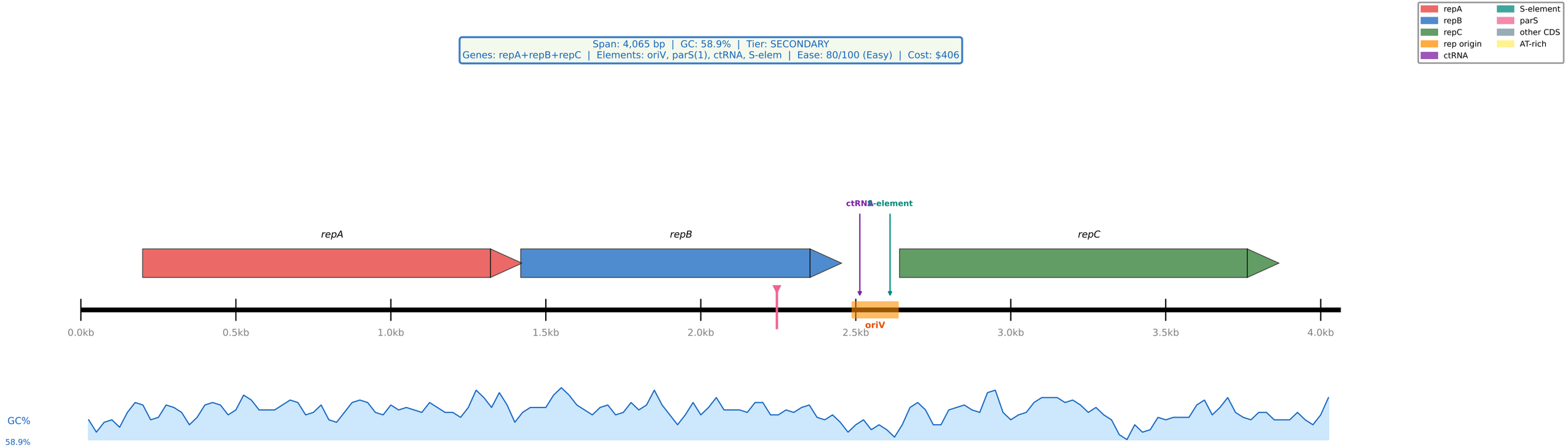
T155_95_NZ_JAAMCX010000007.1 | unknown | Agrobacterium rhizogenes

Span: 3,983 bp | GC: 56.8% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: oriV, parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $398



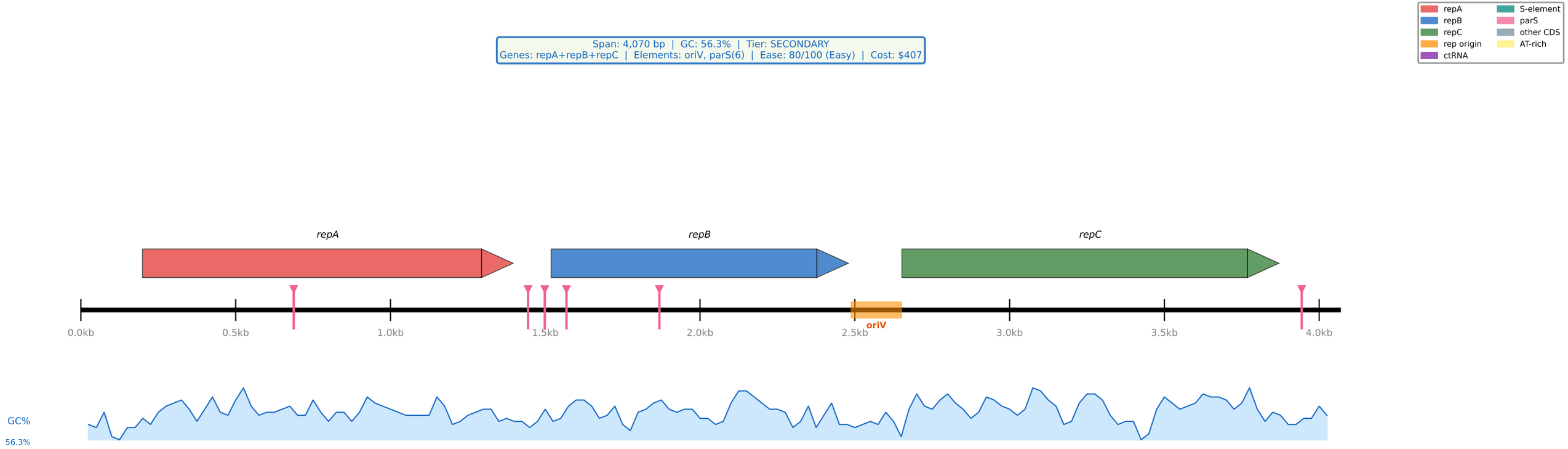
AB2_73_contig_3 | unknown | Agrobacterium tumefaciens

Span: 4,065 bp | GC: 58.9% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: oriV, parS(1), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $406



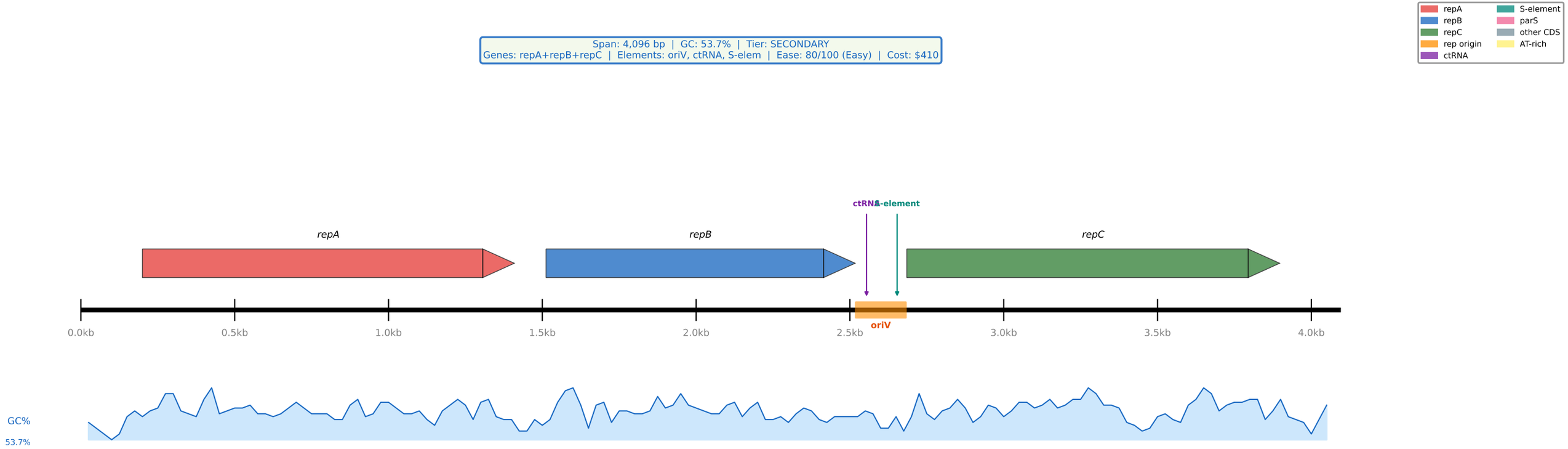
T155_95_NZ_JAAMCX010000013.1 | unknown | Agrobacterium rhizogenes

Span: 4,070 bp | GC: 56.3% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: oriV, parS(6) | Ease: 80/100 (Easy) | Cost: $407



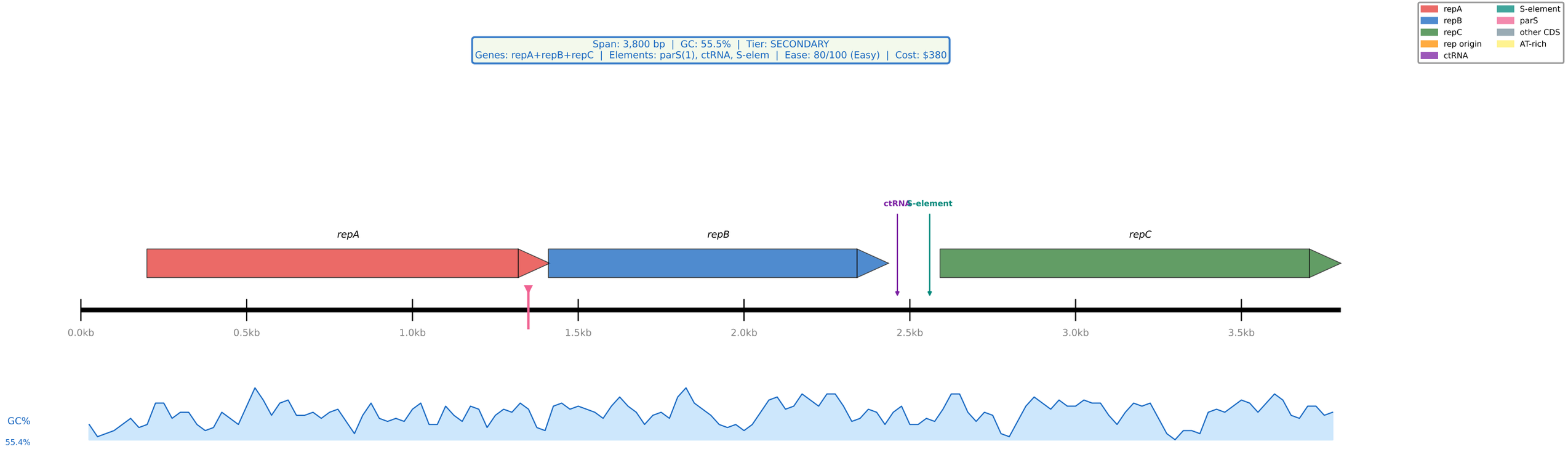
B230_85_contig_4 | unknown | Agrobacterium tumefaciens

Span: 4,096 bp | GC: 53.7% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: oriV, ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $410



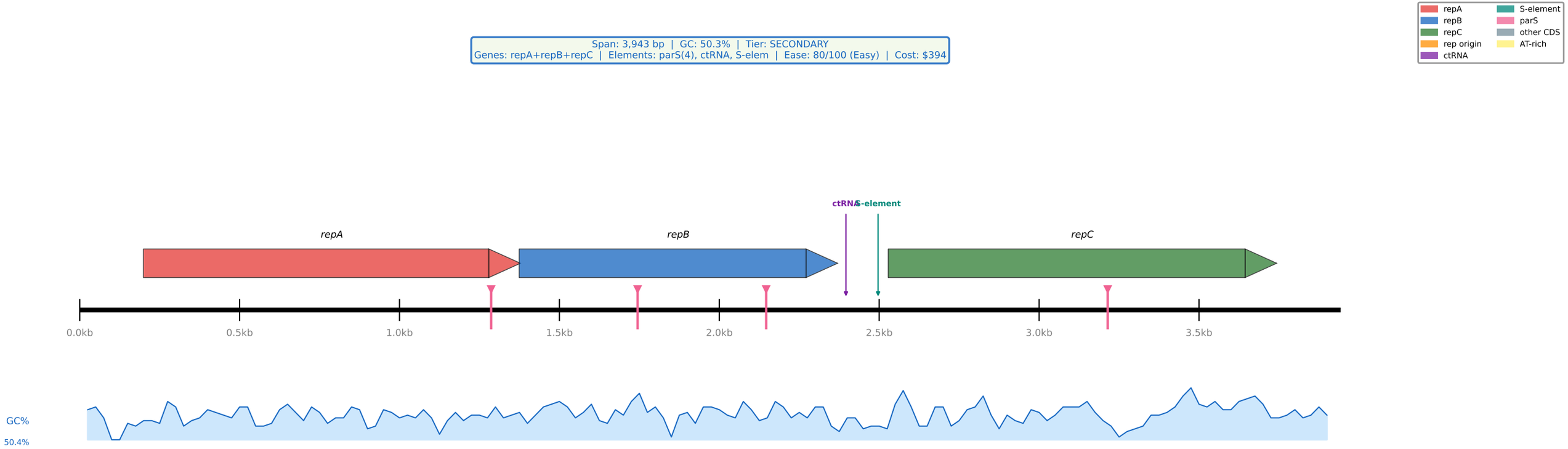
Rhizobium_etli_CFN_42_U80928.5 | unknown | Rhizobium etli CFN 42

Span: 3,800 bp | GC: 55.5% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(1), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $380



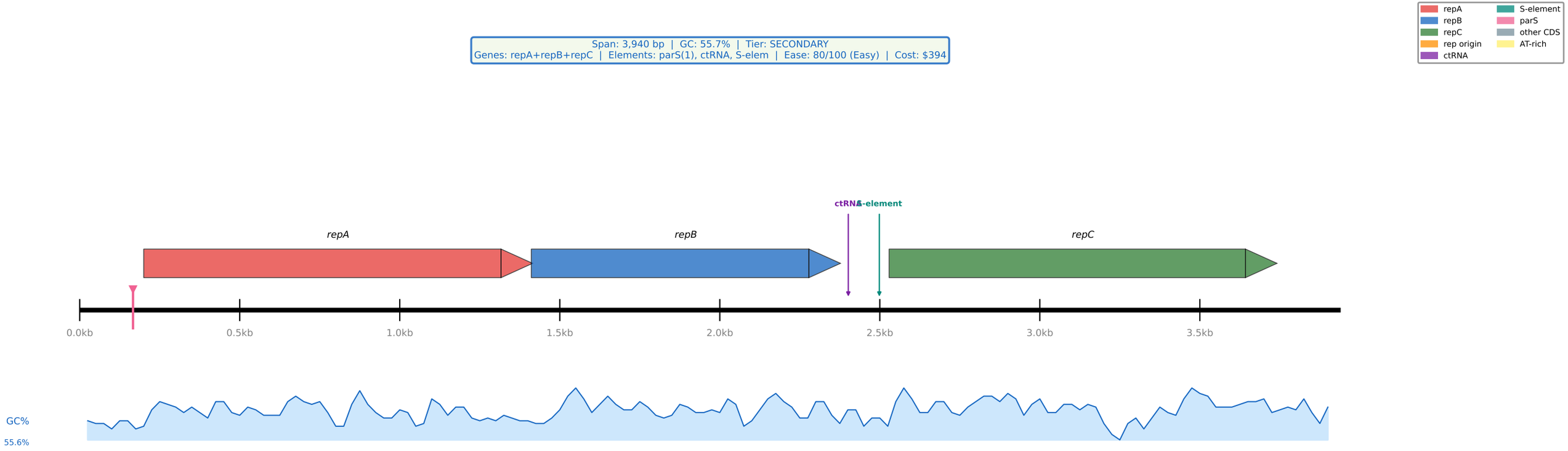
CG507_CG507_8 | unknown | Agrobacterium tumefaciens

Span: 3,943 bp | GC: 50.3% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(4), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $394



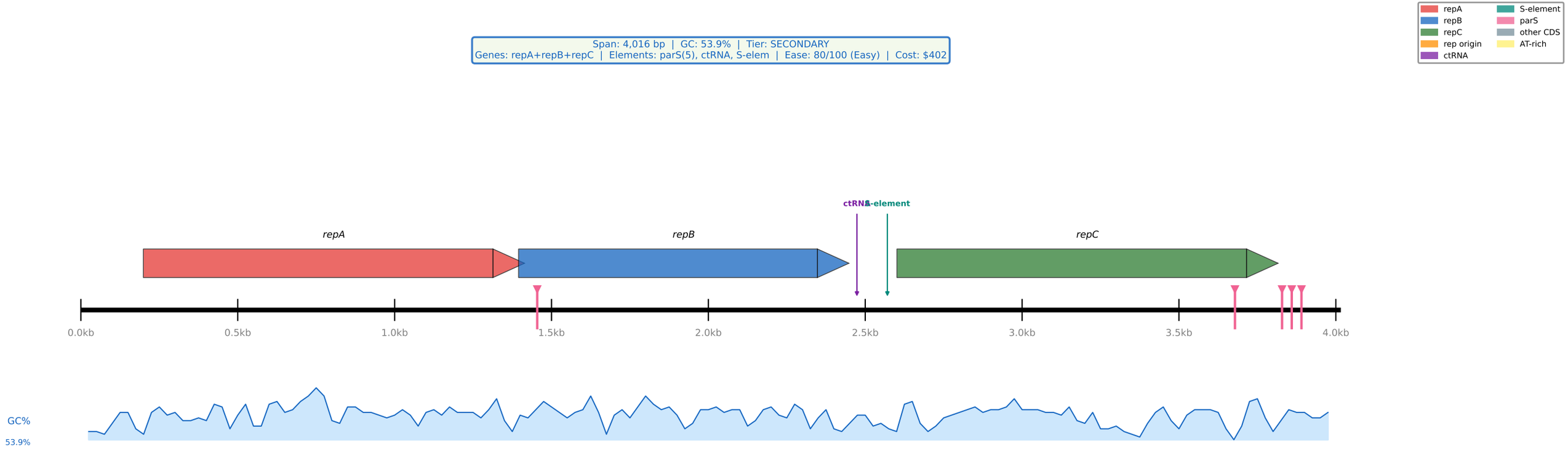
Q15_94_contig_4 | unknown | Agrobacterium tumefaciens

Span: 3,940 bp | GC: 55.7% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(1), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $394



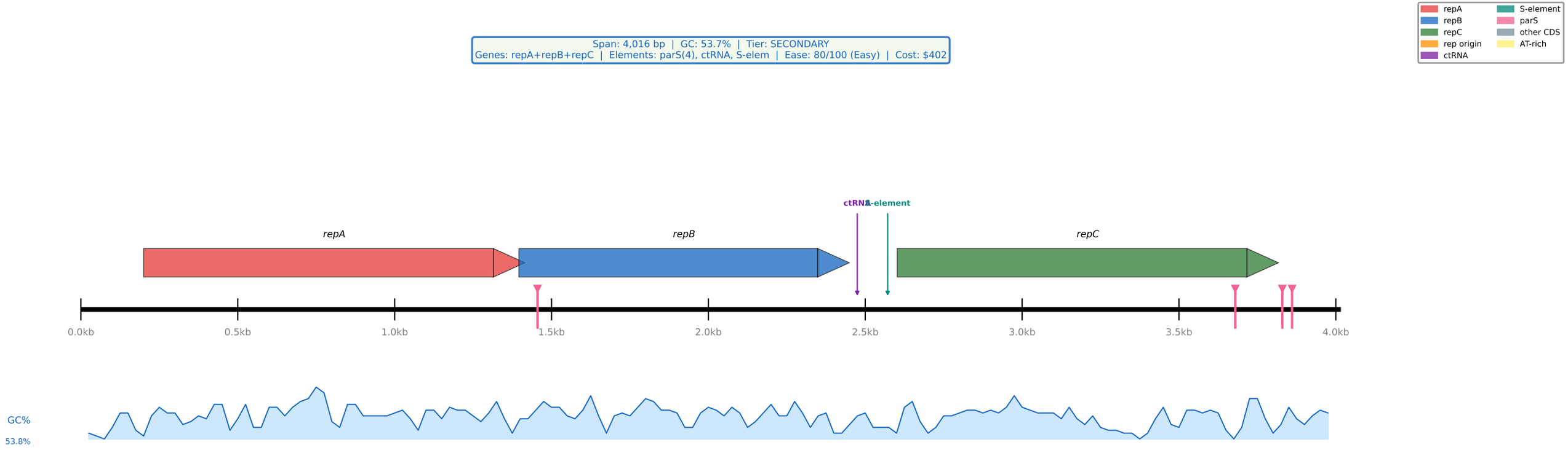
L51_94_contig_2_rep_1 | unknown | Agrobacterium tumefaciens

Span: 4,016 bp | GC: 53.9% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(5), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $402



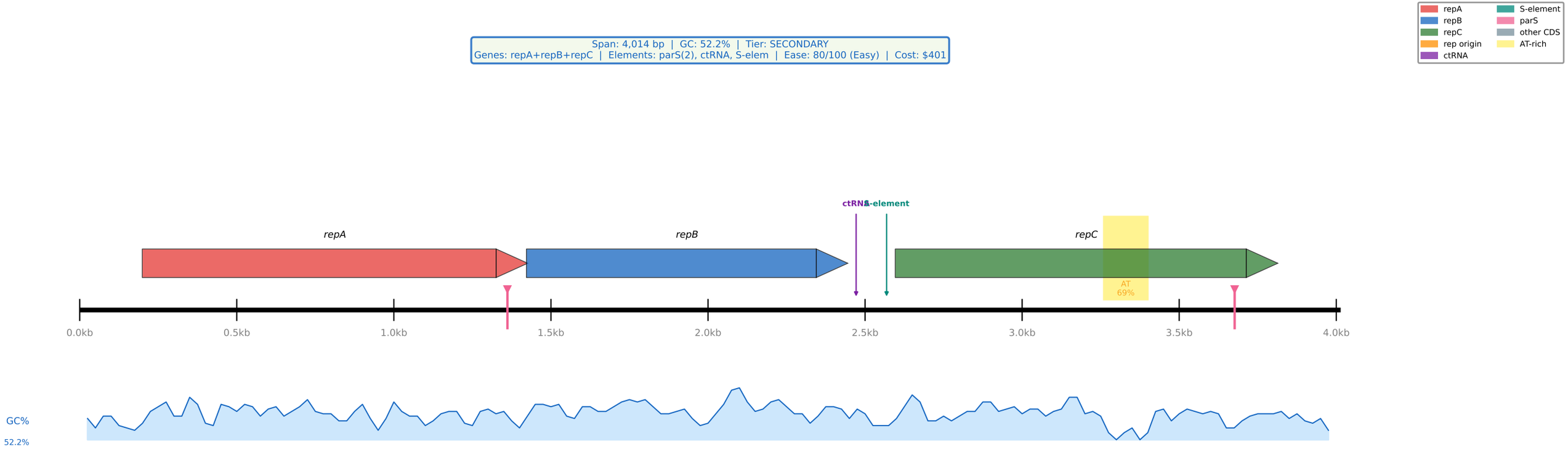
B21_90_contig_2_rep_2 | unknown | Agrobacterium tumefaciens

Span: 4,016 bp | GC: 53.7% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(4), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $402



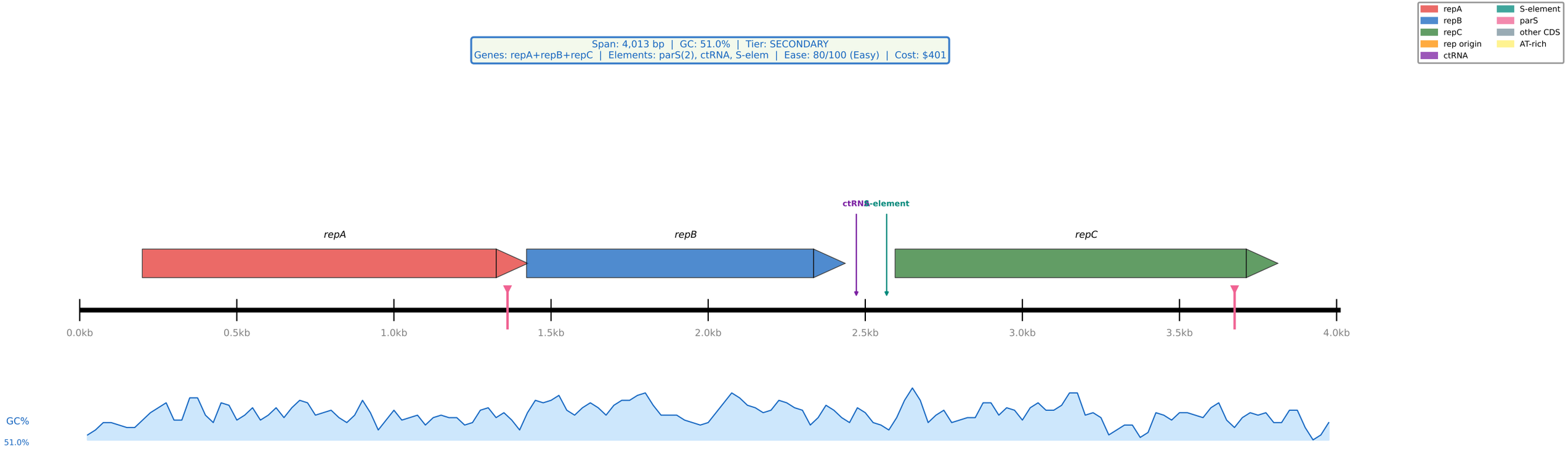
CG958_CG958_3 | unknown | Agrobacterium tumefaciens

Span: 4,014 bp | GC: 52.2% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $401



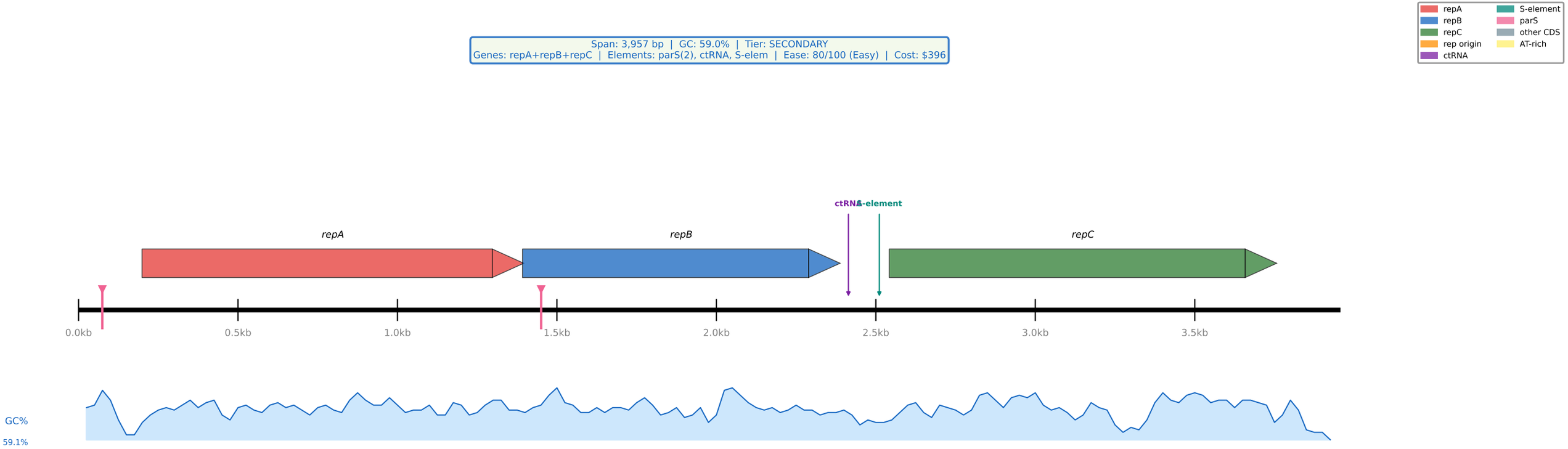
AMS008_pAt | pAt | Agrobacterium tumefaciens

Span: 4,013 bp | GC: 51.0% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $401



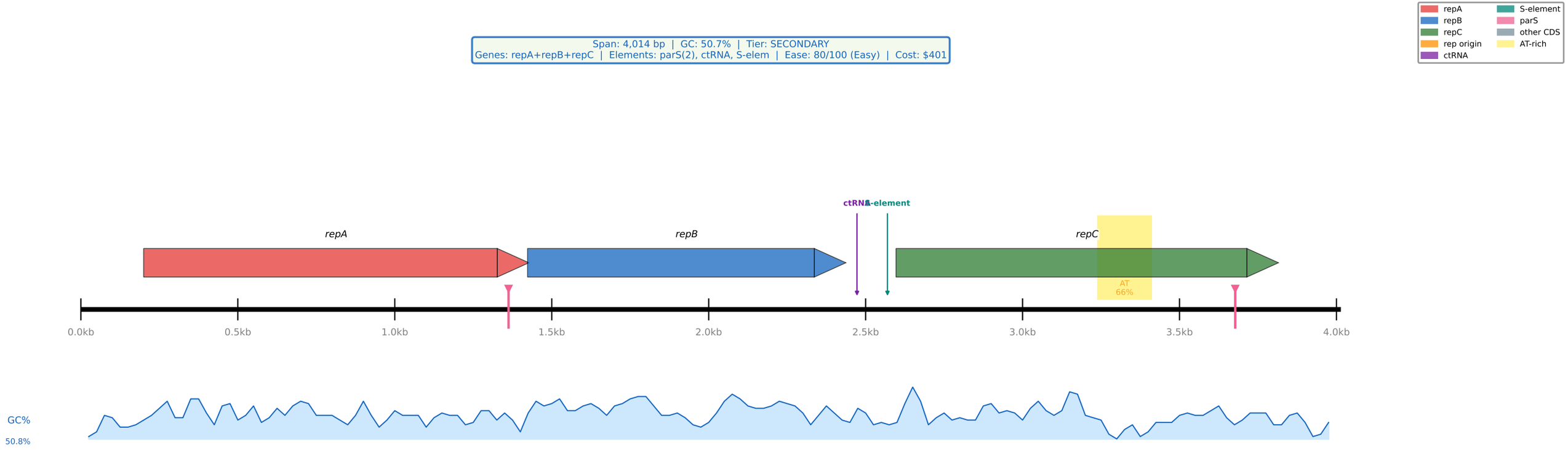
WS16_contig_4 | unknown | Agrobacterium tumefaciens

Span: 3,957 bp | GC: 59.0% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $396



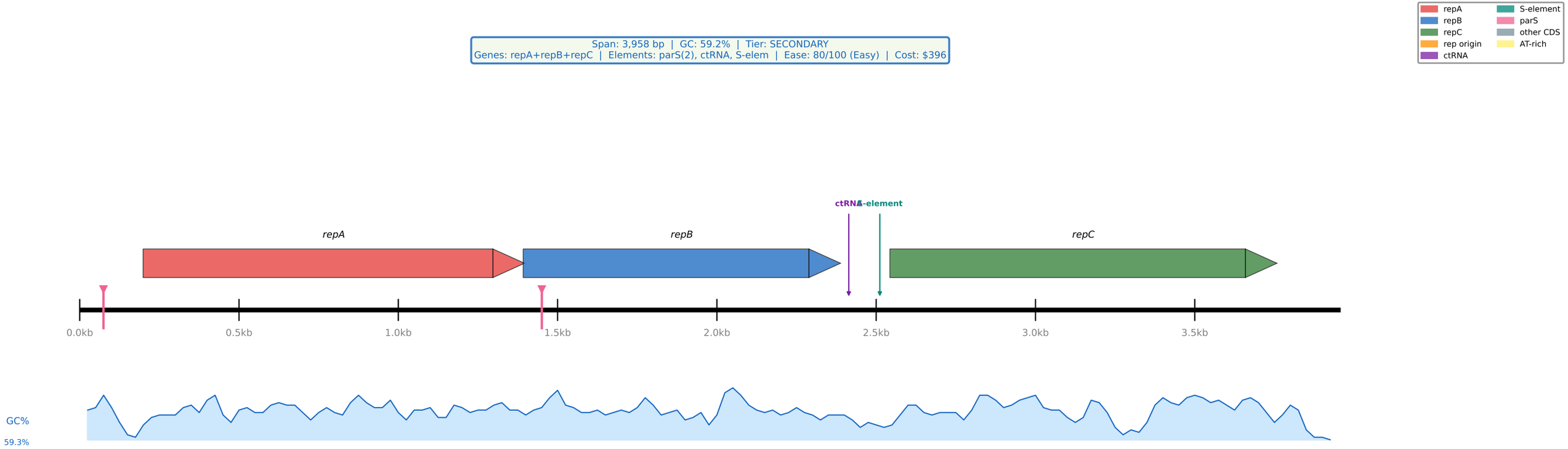
1D1514_pAt | pAt | Agrobacterium tumefaciens

Span: 4,014 bp | GC: 50.7% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $401



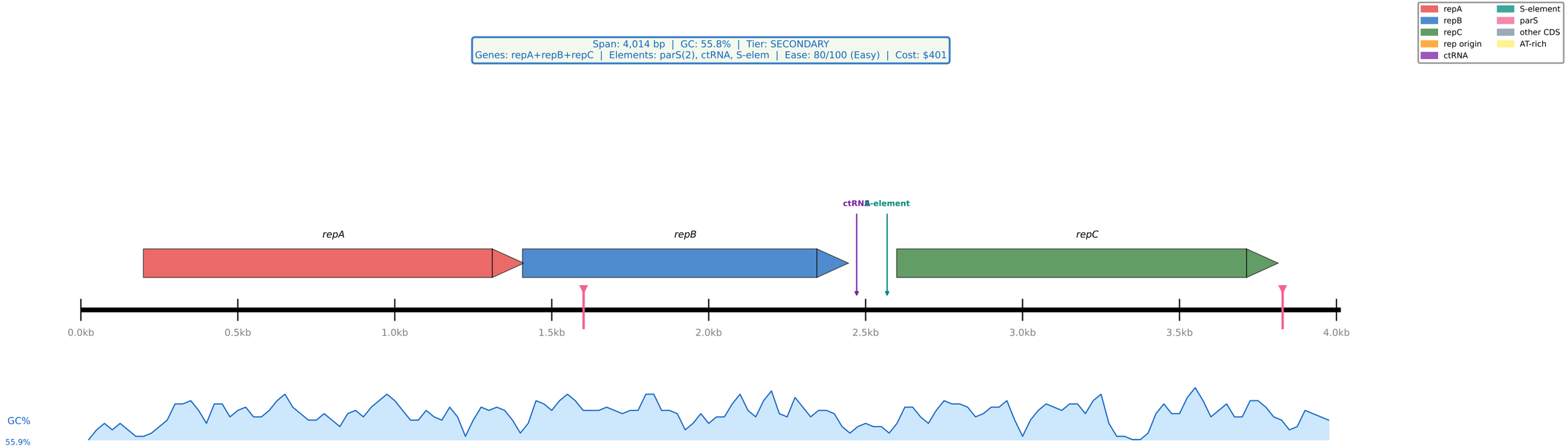
WS192_contig_4 | unknown | Agrobacterium tumefaciens

Span: 3,958 bp | GC: 59.2% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $396



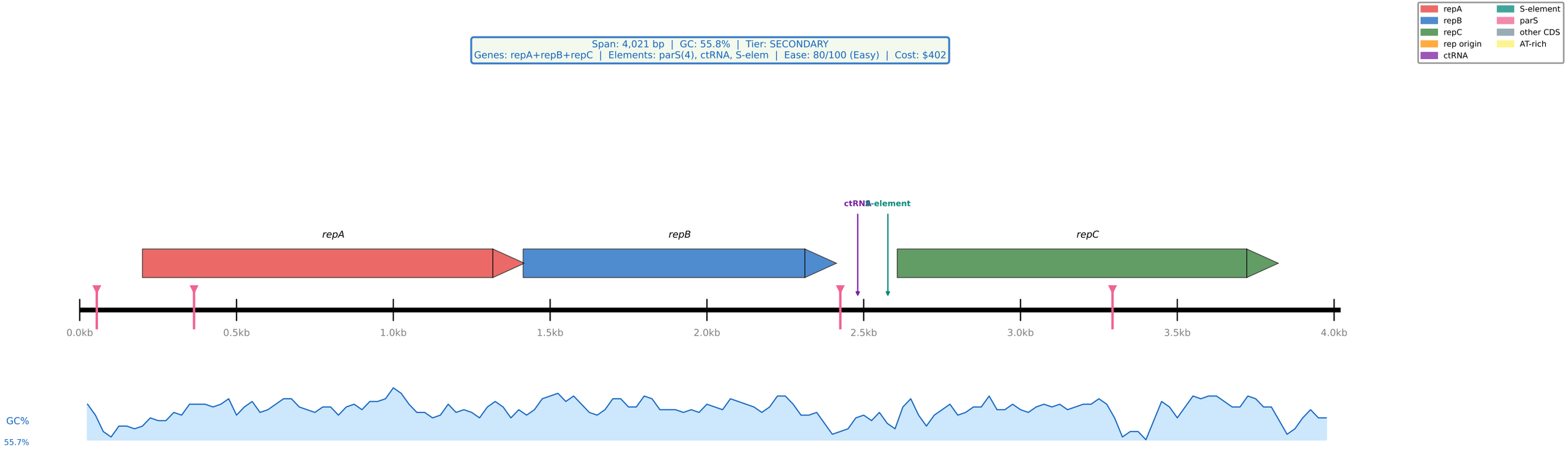
WS287_contig_6_rep_2 | unknown | Agrobacterium tumefaciens

Span: 4,014 bp | GC: 55.8% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $401



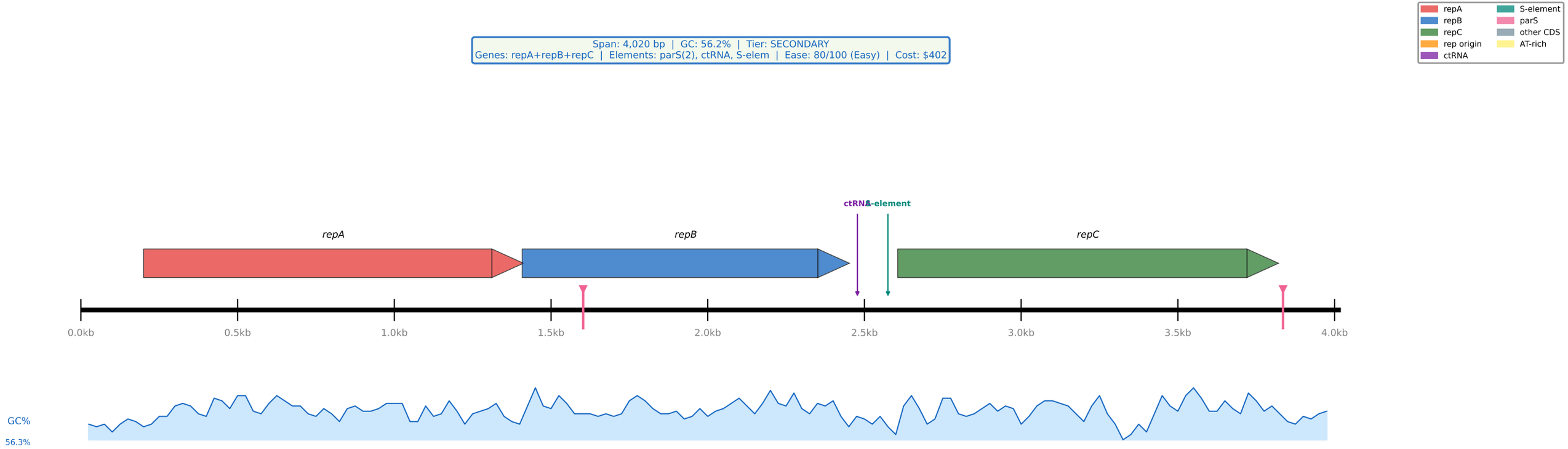
Q15_94_contig_7 | unknown | Agrobacterium tumefaciens

Span: 4,021 bp | GC: 55.8% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(4), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $402



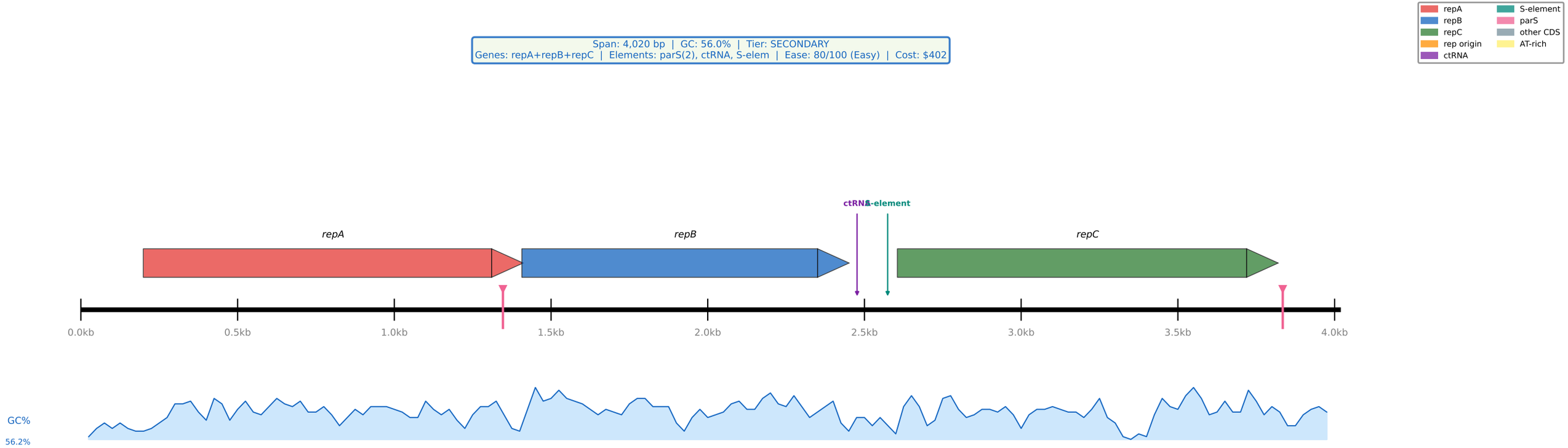
WS163_contig_7 | unknown | Agrobacterium tumefaciens

Span: 4,020 bp | GC: 56.2% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $402



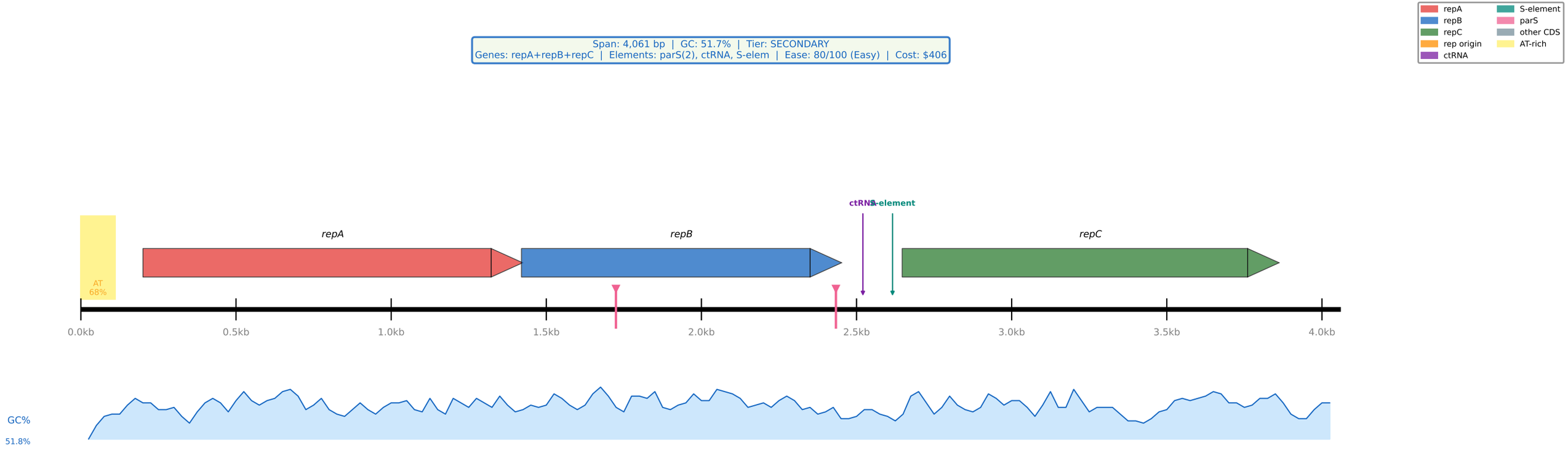
WS16_contig_2 | unknown | Agrobacterium tumefaciens

Span: 4,020 bp | GC: 56.0% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $402



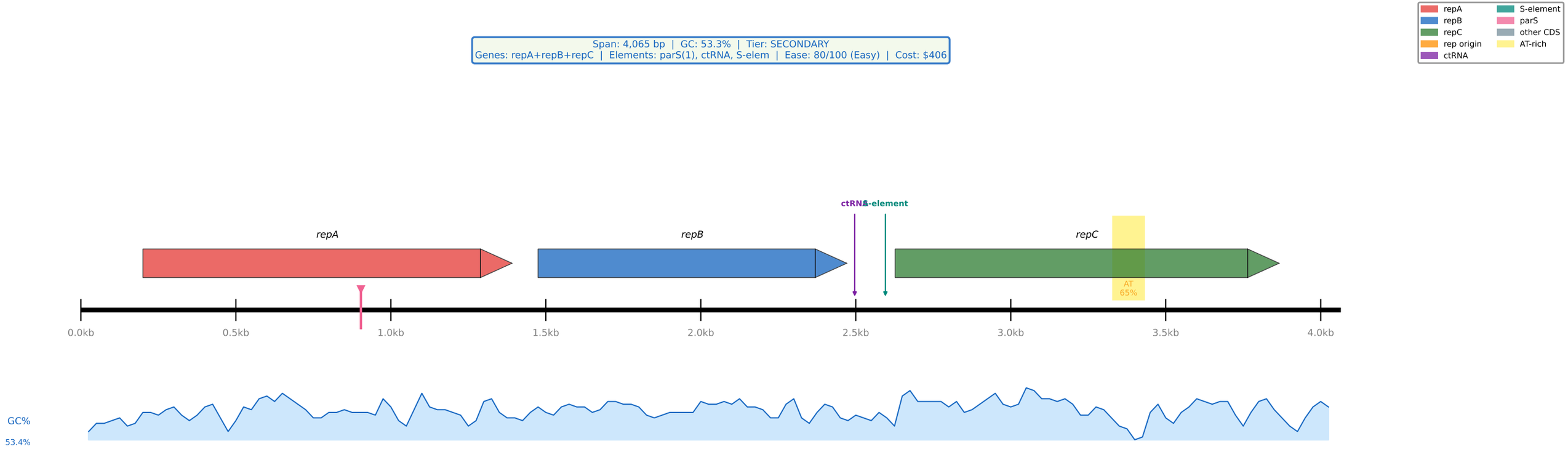
CG507_CG507_2 | unknown | Agrobacterium tumefaciens

Span: 4,061 bp | GC: 51.7% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $406



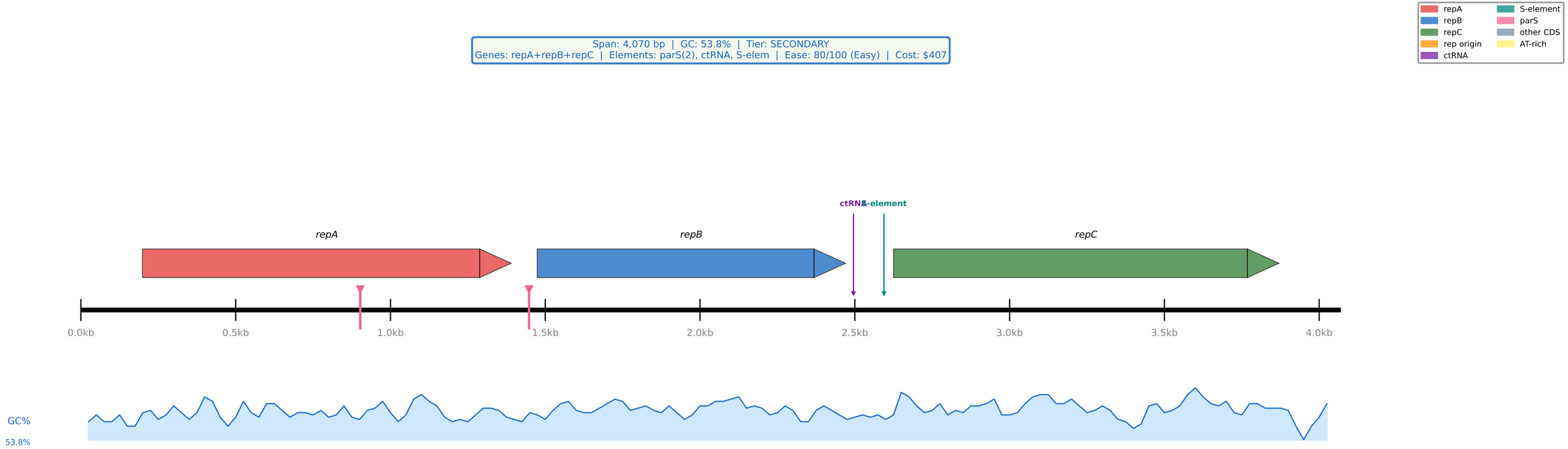
1D1524_plasmid1 | unknown | Agrobacterium tumefaciens

Span: 4,065 bp | GC: 53.3% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(1), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $406



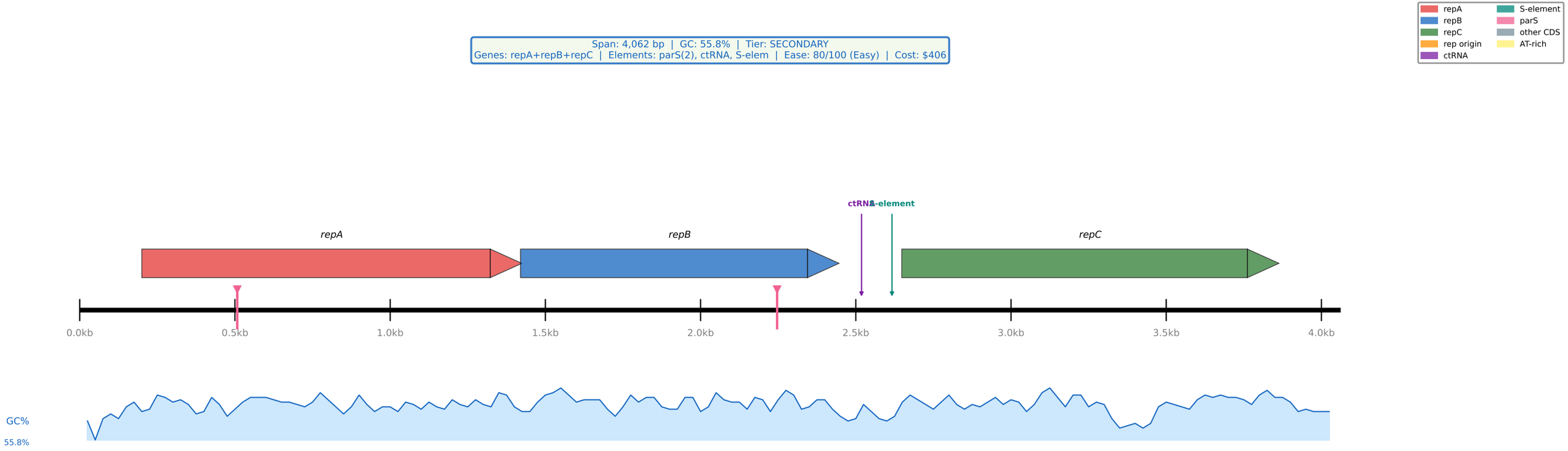
CG474_CG474_4_rep_2 | unknown | Agrobacterium tumefaciens

Span: 4,070 bp | GC: 53.8% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $407



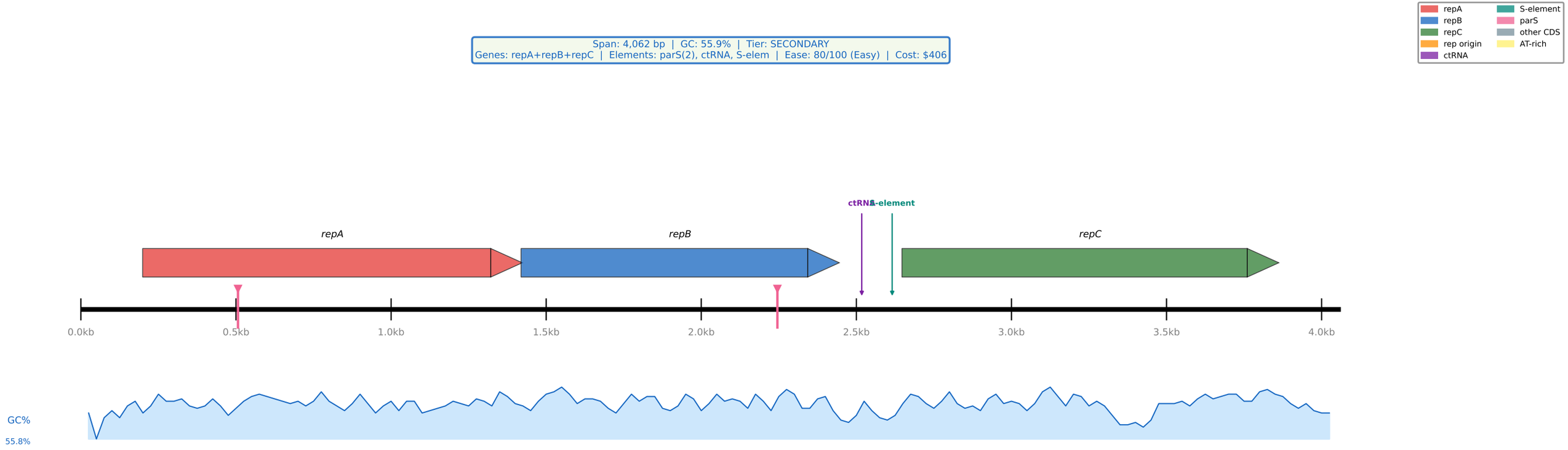
Q15_94_contig_3 | unknown | Agrobacterium tumefaciens

Span: 4,062 bp | GC: 55.8% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $406



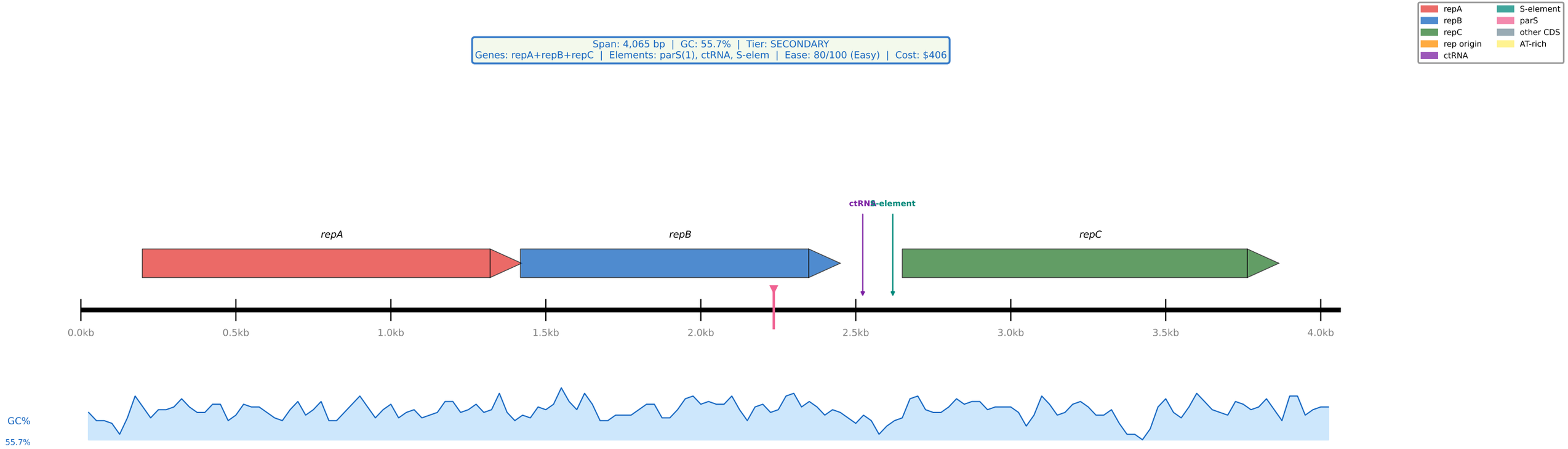
S7_73_contig_4 | unknown | Agrobacterium tumefaciens

Span: 4,062 bp | GC: 55.9% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $406



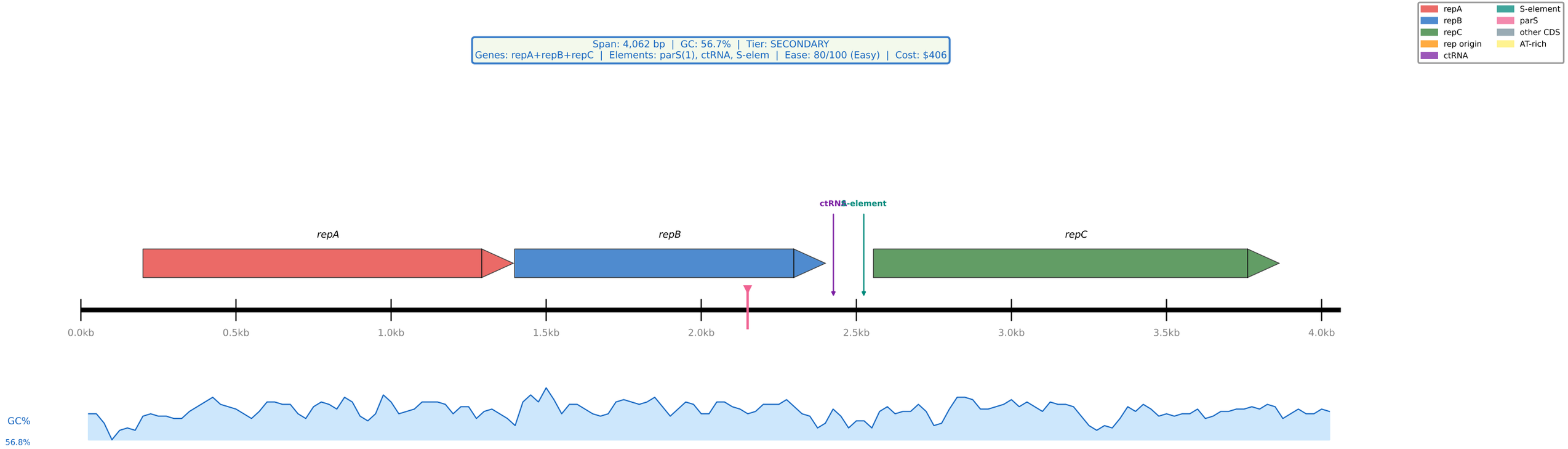
C58_AE007872.2 | unknown | Agrobacterium fabrum

Span: 4,065 bp | GC: 55.7% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(1), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $406



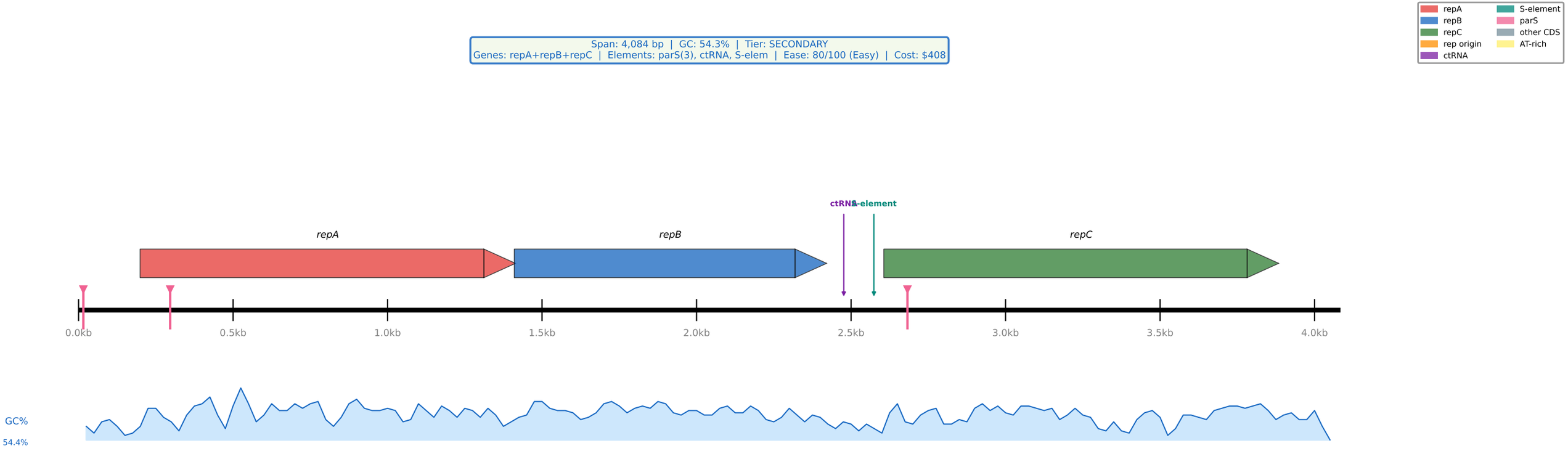
AB2_73_contig_2 | unknown | Agrobacterium tumefaciens

Span: 4,062 bp | GC: 56.7% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(1), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $406



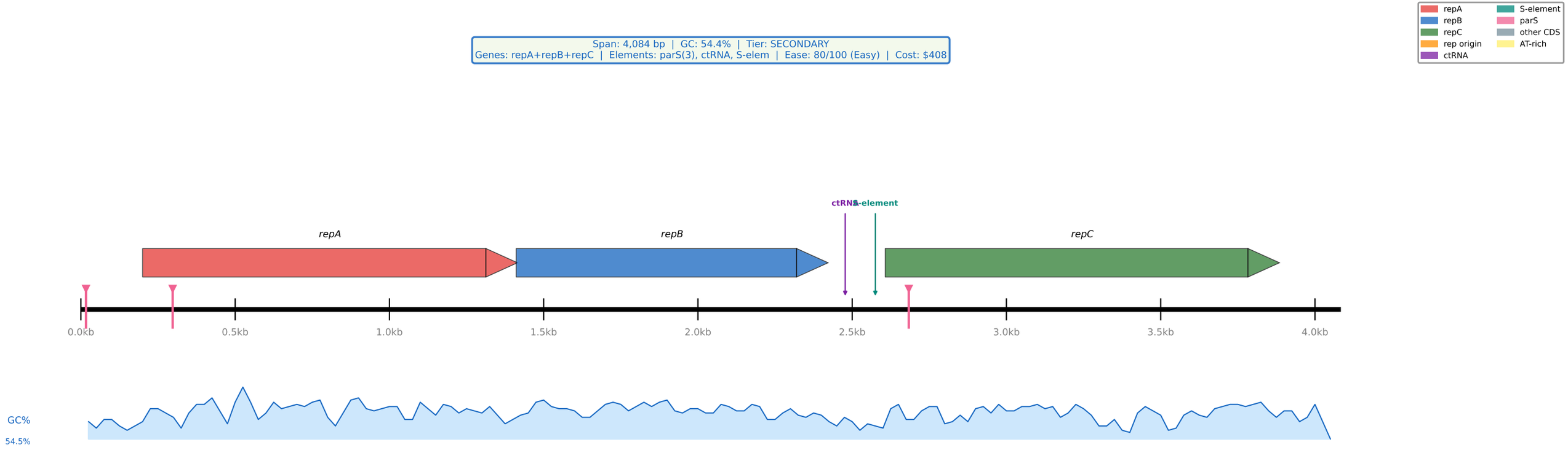
Q15_94_contig_2 | unknown | Agrobacterium tumefaciens

Span: 4,084 bp | GC: 54.3% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(3), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $408



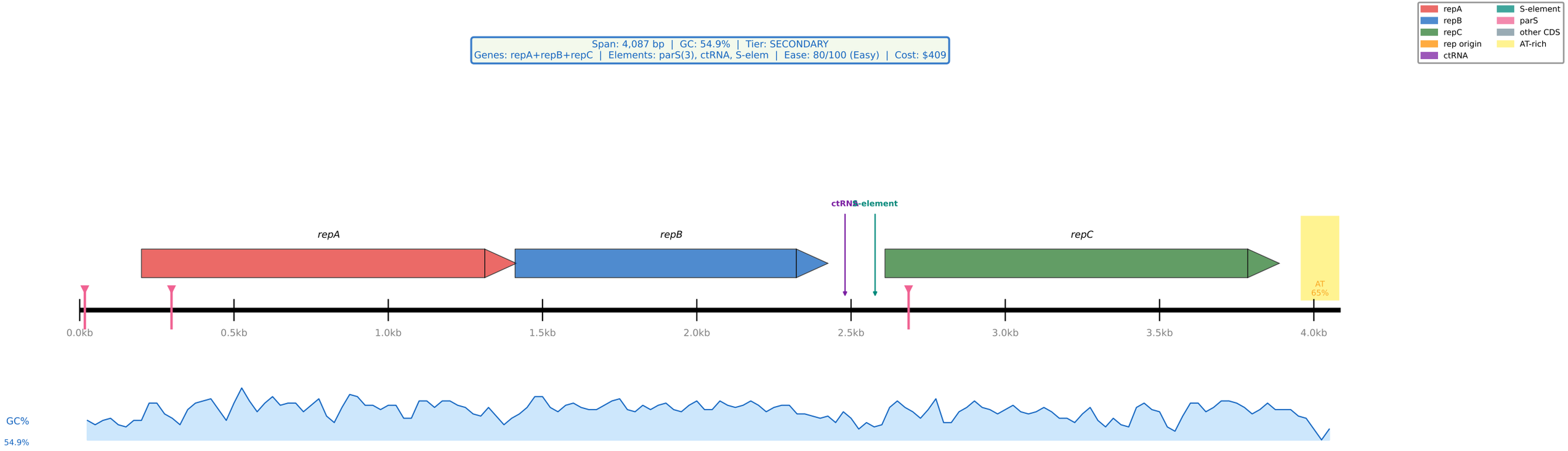
15-1187-1-2a_CP123839.1 | unknown | Agrobacterium tumefaciens

Span: 4,084 bp | GC: 54.4% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(3), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $408



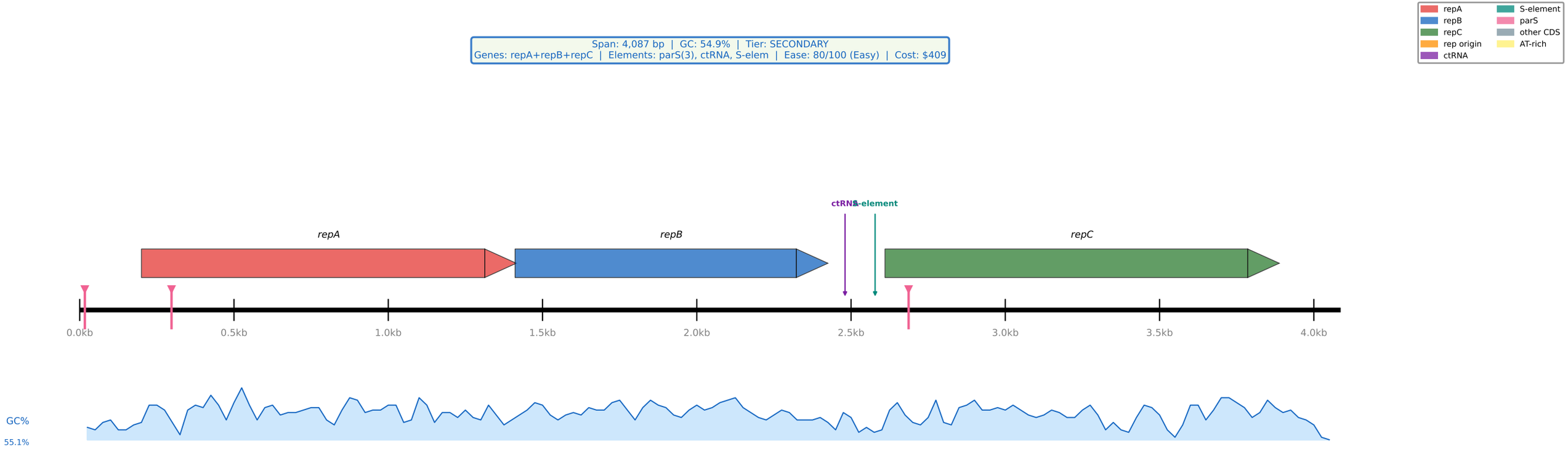
C58_AE007870.2 | unknown | Agrobacterium fabrum

Span: 4,087 bp | GC: 54.9% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(3), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $409



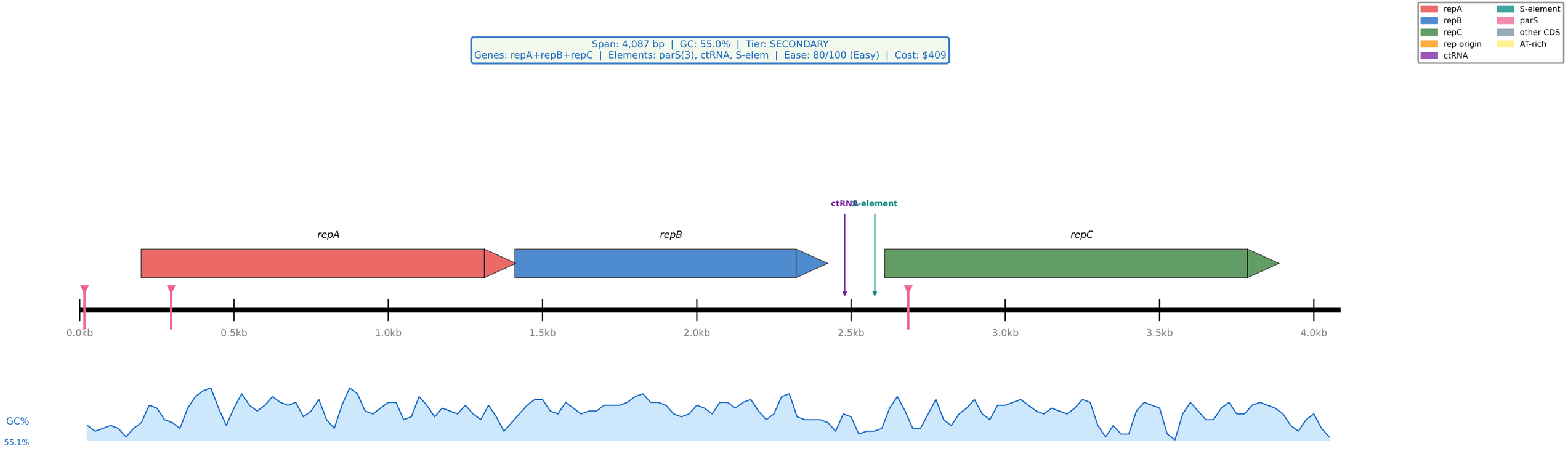
K599_CAICSX020000002.1 | unknown | Agrobacterium rhizogenes

Span: 4,087 bp | GC: 54.9% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(3), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $409



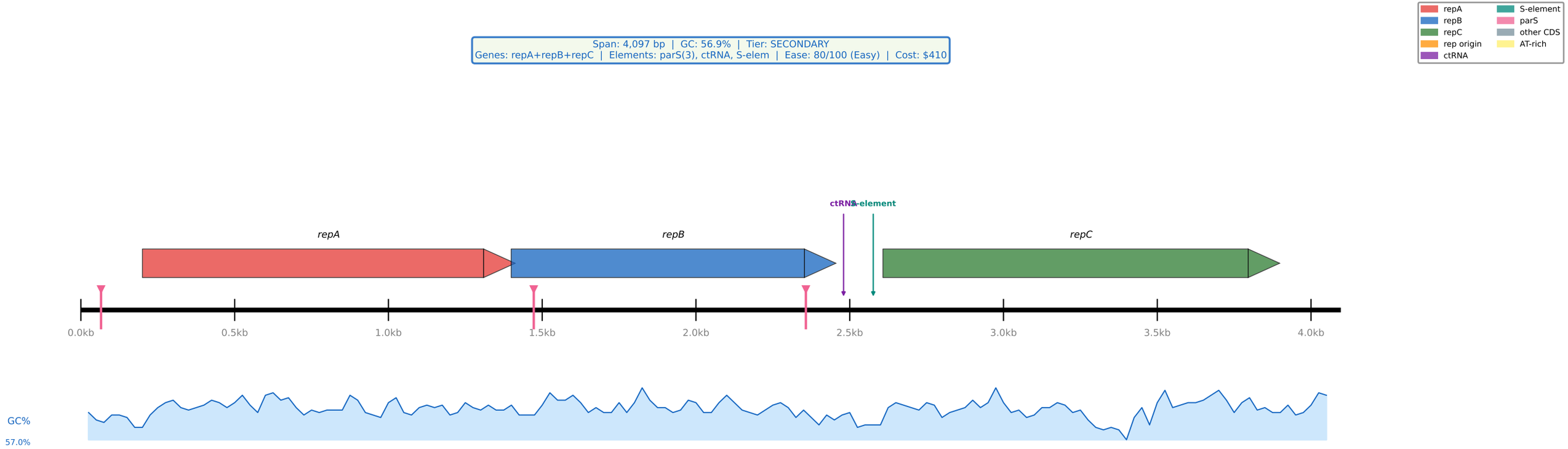
NCPPB_1641_NZ_LT009776.1 | unknown | Agrobacterium vitis

Span: 4,087 bp | GC: 55.0% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(3), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $409



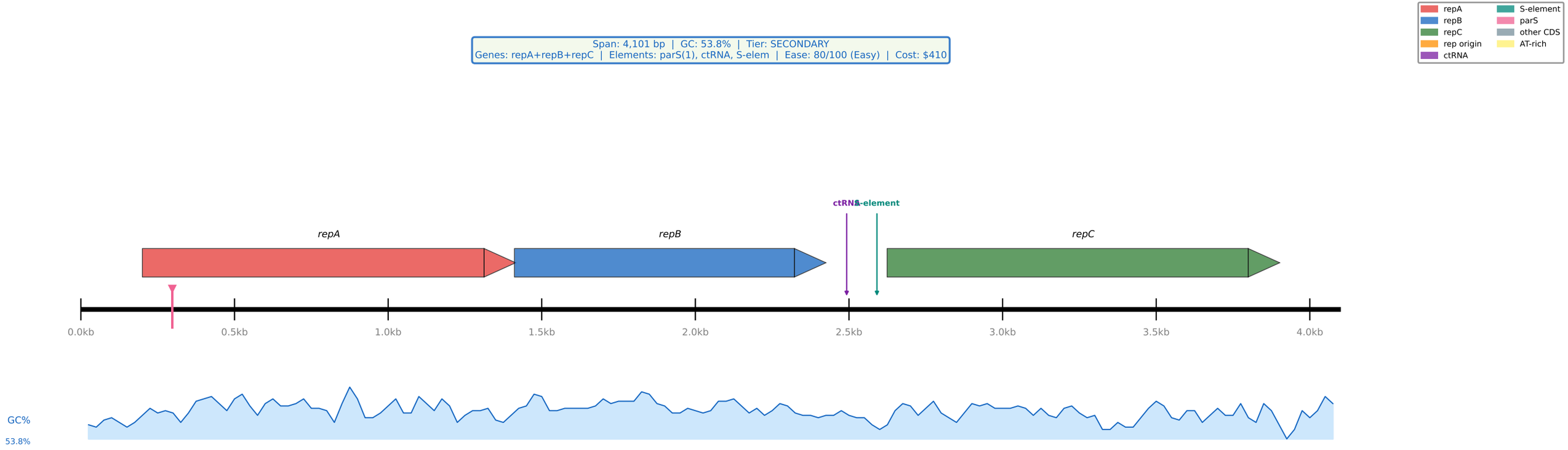
L51_94_contig_4 | unknown | Agrobacterium tumefaciens

Span: 4,097 bp | GC: 56.9% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(3), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $410



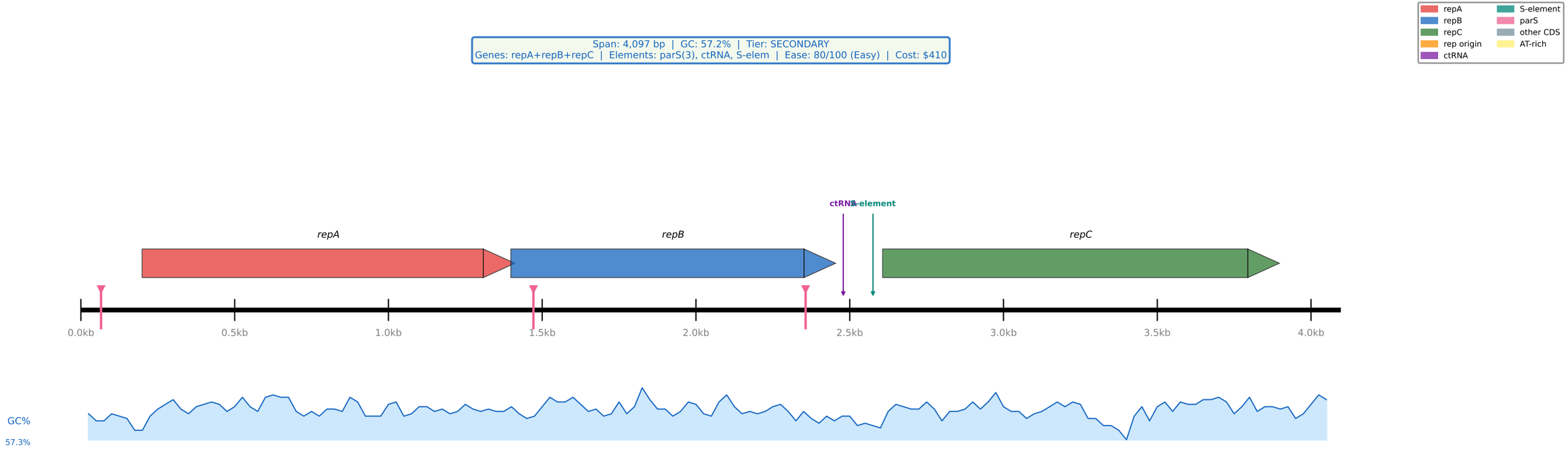
AF3.44_contig_2 | unknown | Agrobacterium tumefaciens

Span: 4,101 bp | GC: 53.8% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(1), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $410



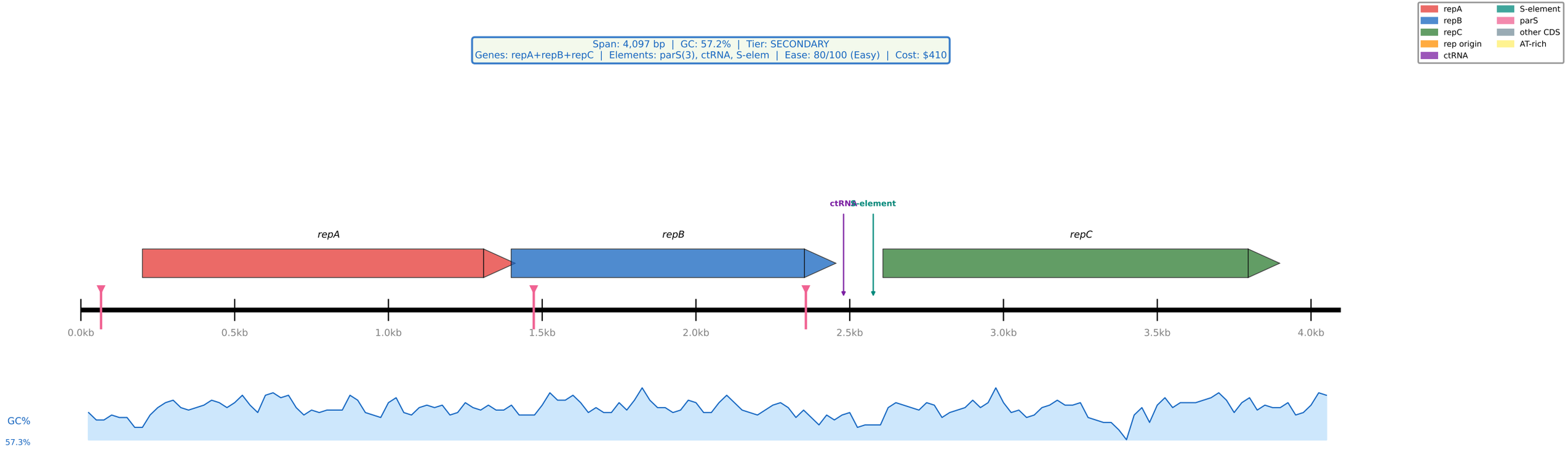
B230_85_contig_3 | unknown | Agrobacterium tumefaciens

Span: 4,097 bp | GC: 57.2% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(3), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $410



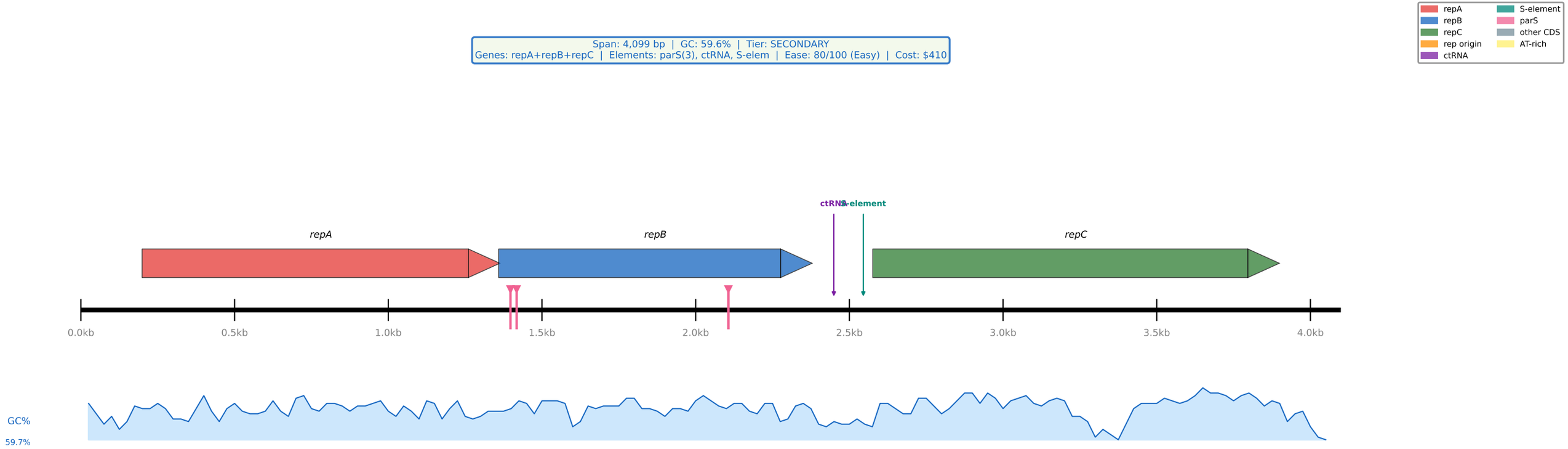
B21_90_contig_4 | unknown | Agrobacterium tumefaciens

Span: 4,097 bp | GC: 57.2% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(3), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $410



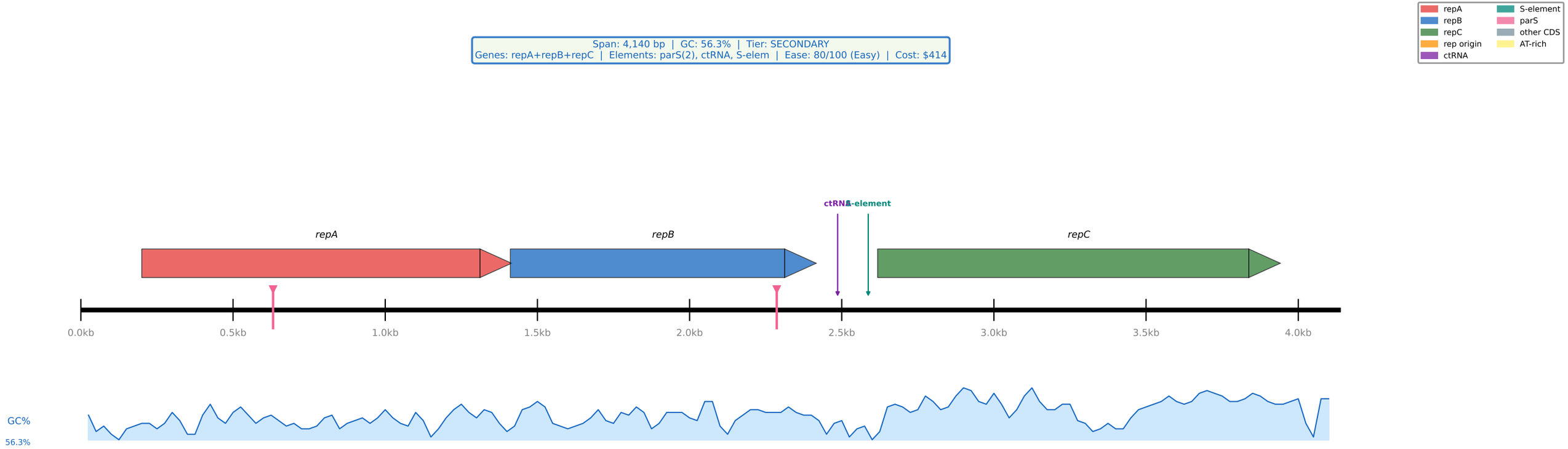
L51_94_contig_2_rep_2 | unknown | Agrobacterium tumefaciens

Span: 4,099 bp | GC: 59.6% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(3), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $410



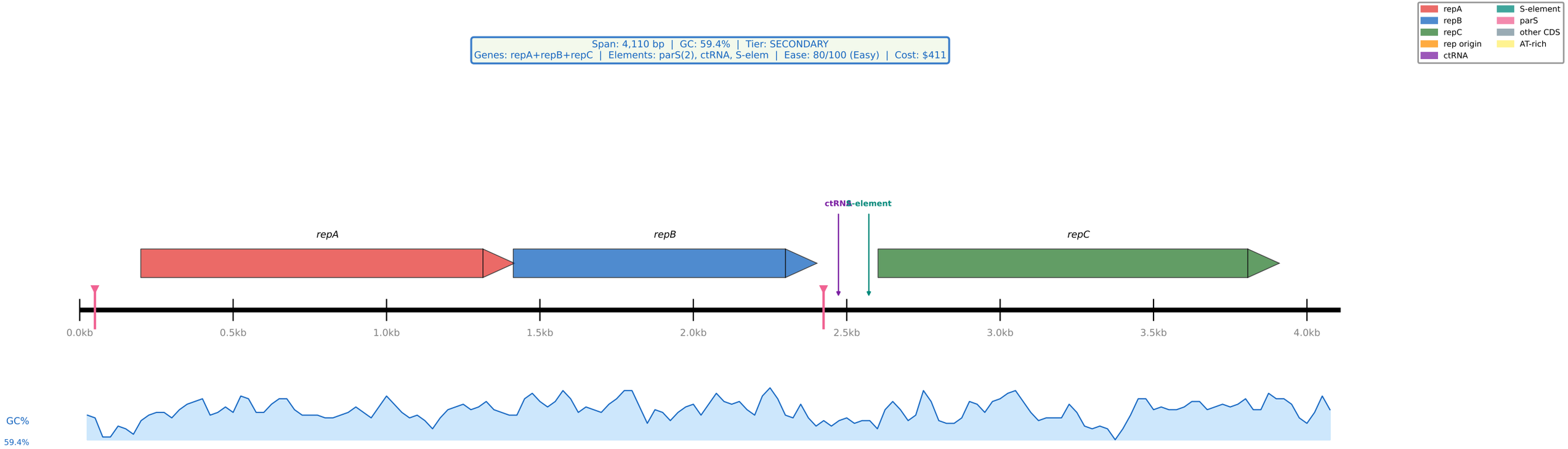
L51_94_contig_3 | unknown | Agrobacterium tumefaciens

Span: 4,140 bp | GC: 56.3% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $414



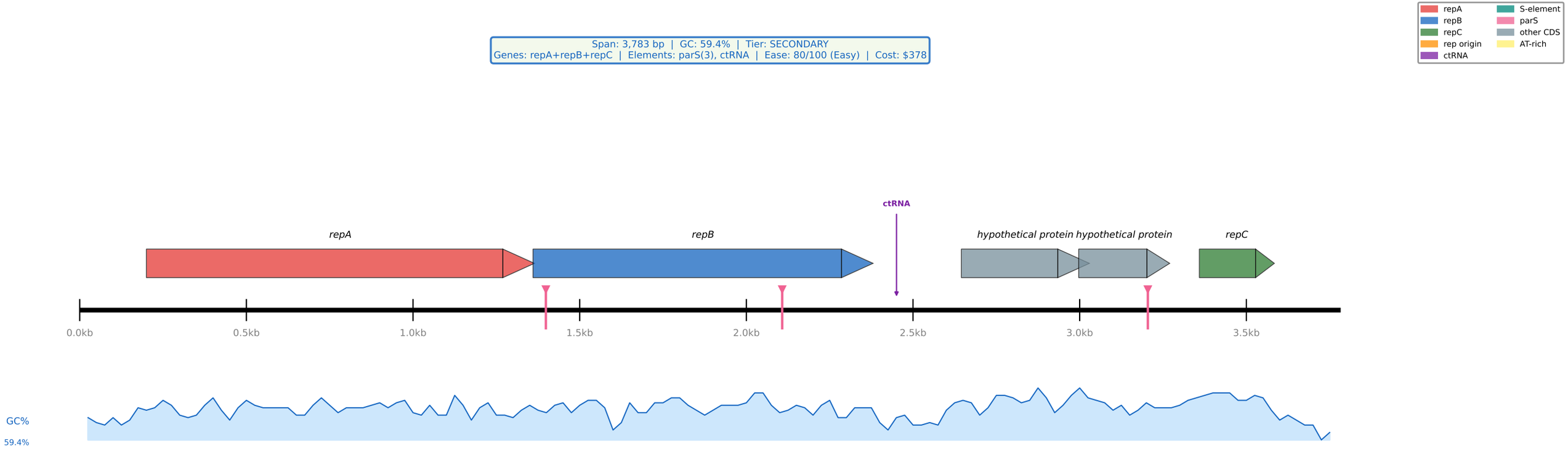
WS16_contig_5 | unknown | Agrobacterium tumefaciens

Span: 4,110 bp | GC: 59.4% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $411



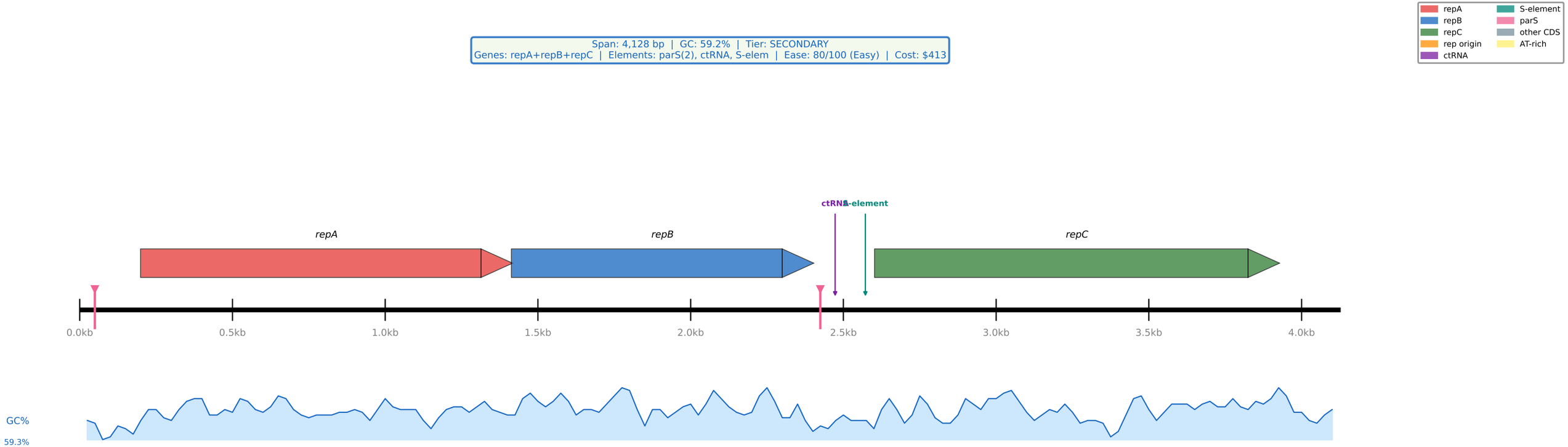
B21_90_contig_2_rep_1 | unknown | Agrobacterium tumefaciens

Span: 3,783 bp | GC: 59.4% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(3), ctRNA | Ease: 80/100 (Easy) | Cost: $378



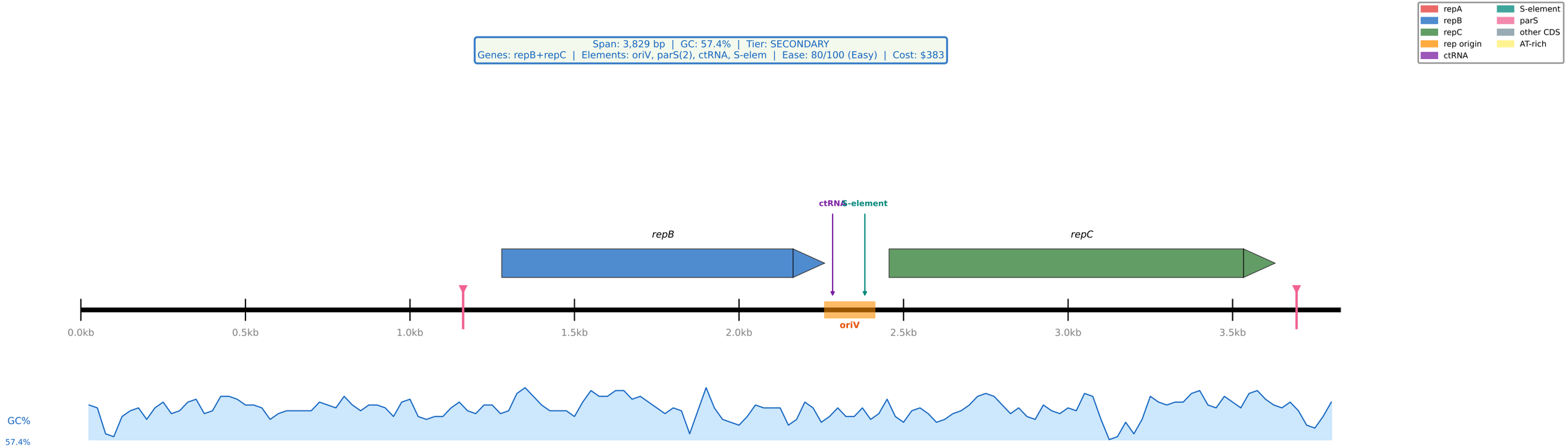
WS192_contig_2 | unknown | Agrobacterium tumefaciens

Span: 4,128 bp | GC: 59.2% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $413



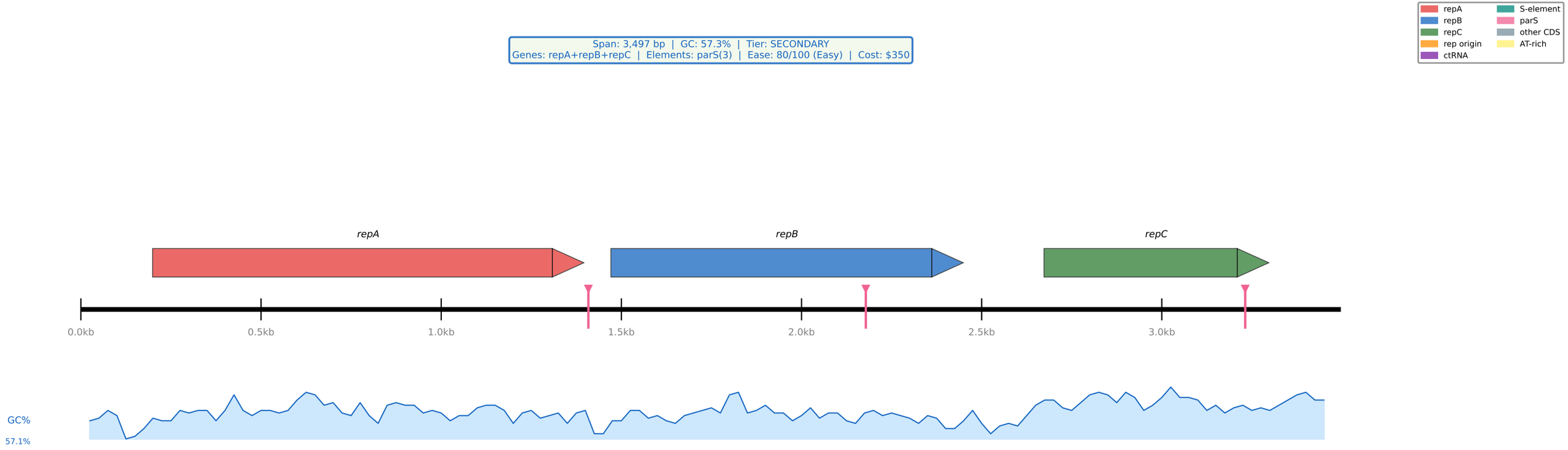
Sinorhizobium_fredii_NGR234_U00090.2 | unknown | Sinorhizobium fredii NGR234

Span: 3,829 bp | GC: 57.4% | Tier: SECONDARY
Genes: repB+repC | Elements: oriV, parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $383



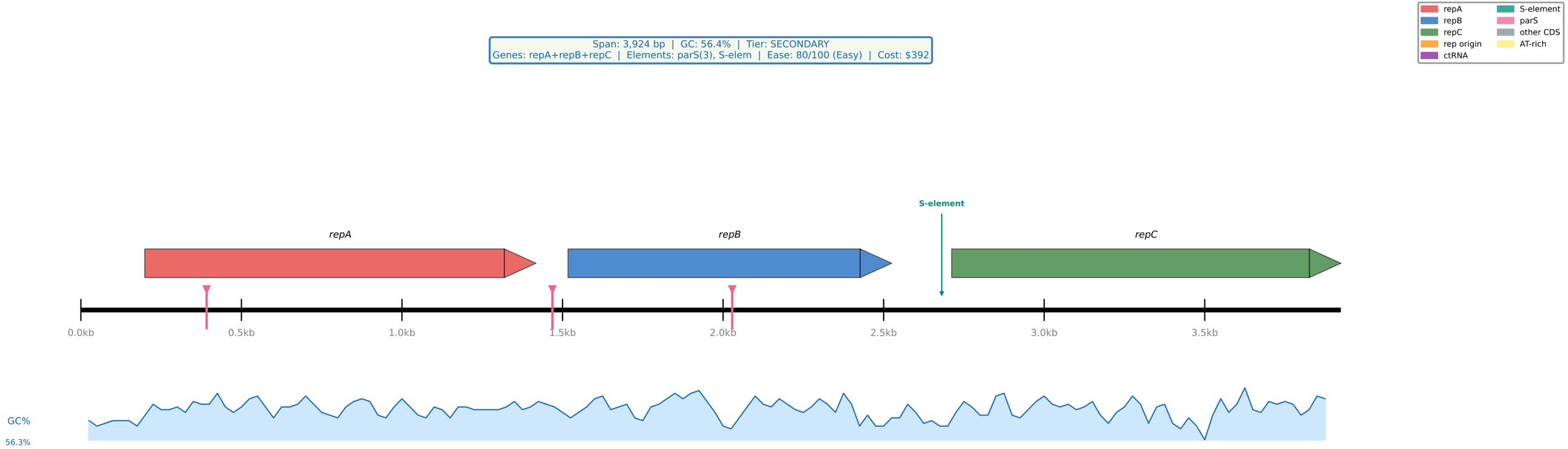
B230_85_contig_2_rep_1 | unknown | Agrobacterium tumefaciens

Span: 3,497 bp | GC: 57.3% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(3) | Ease: 80/100 (Easy) | Cost: $350



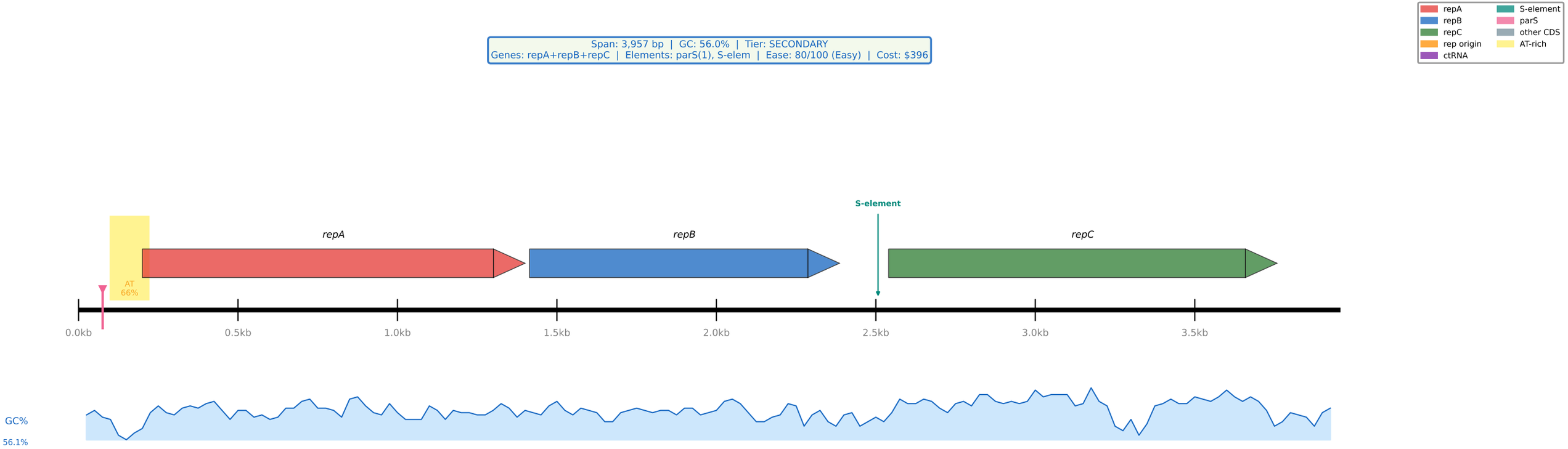
Rhizobium_etli_CFN_42_CP000134.1_rep_2 | unknown | Rhizobium etli CFN 42

Span: 3,924 bp | GC: 56.4% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(3), S-elem | Ease: 80/100 (Easy) | Cost: $392



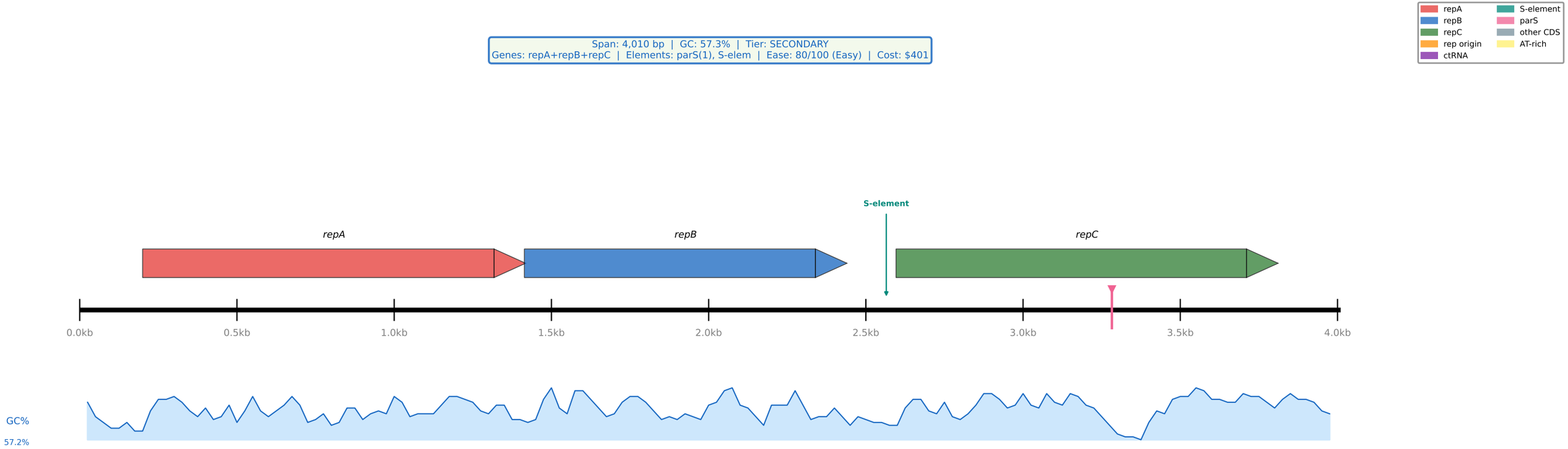
AMS008_plasmid4 | unknown | Agrobacterium tumefaciens

Span: 3,957 bp | GC: 56.0% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(1), S-elem | Ease: 80/100 (Easy) | Cost: $396



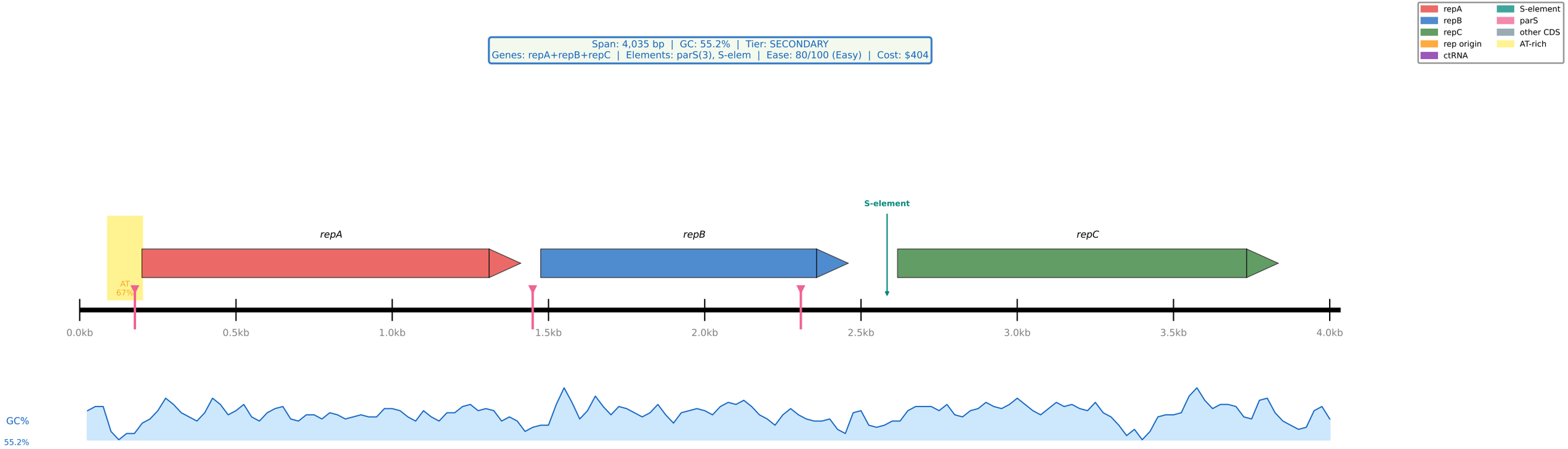
AB2_73_contig_5 | unknown | Agrobacterium tumefaciens

Span: 4,010 bp | GC: 57.3% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(1), S-elem | Ease: 80/100 (Easy) | Cost: $401



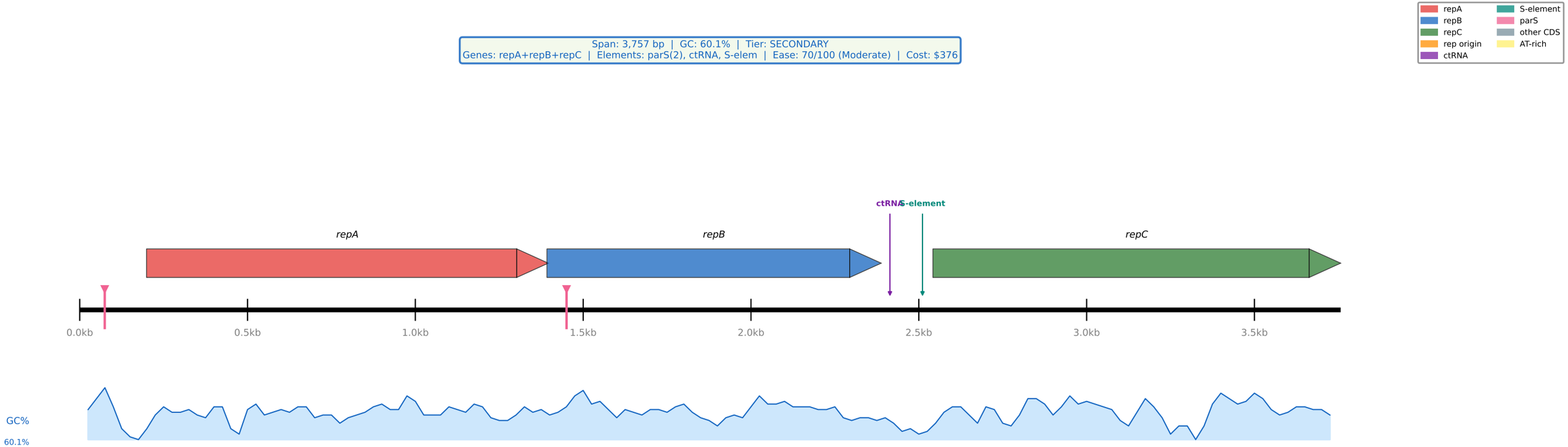
1D1514_plasmid3 | unknown | Agrobacterium tumefaciens

Span: 4,035 bp | GC: 55.2% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(3), S-elem | Ease: 80/100 (Easy) | Cost: $404



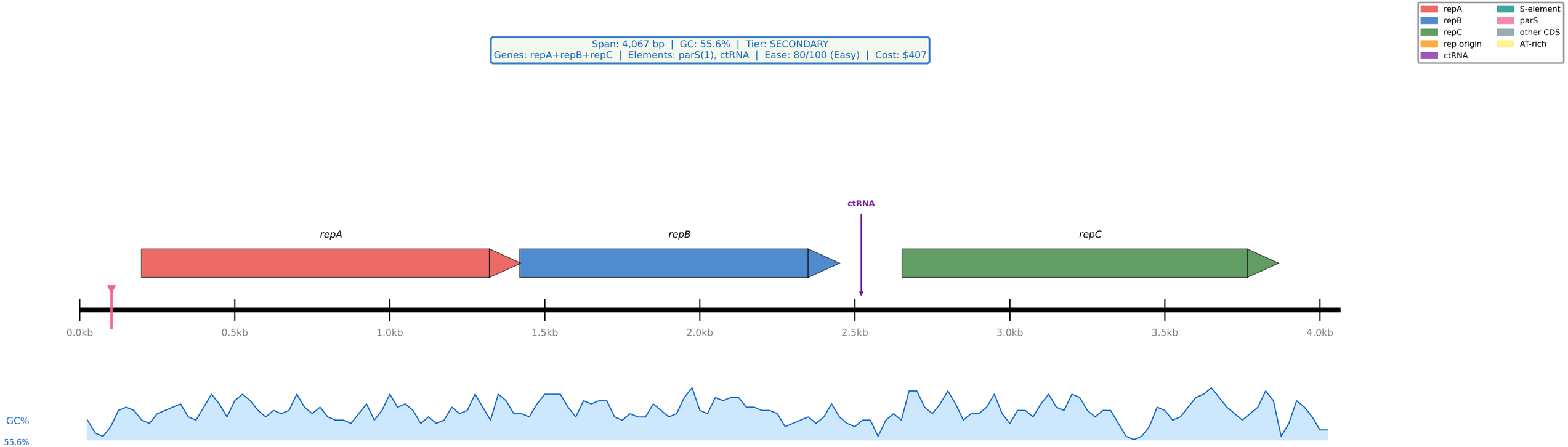
Rhizobium_etli_CFN_42_CP000137.1 | unknown | Rhizobium etli CFN 42

Span: 3,757 bp | GC: 60.1% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 70/100 (Moderate) | Cost: $376



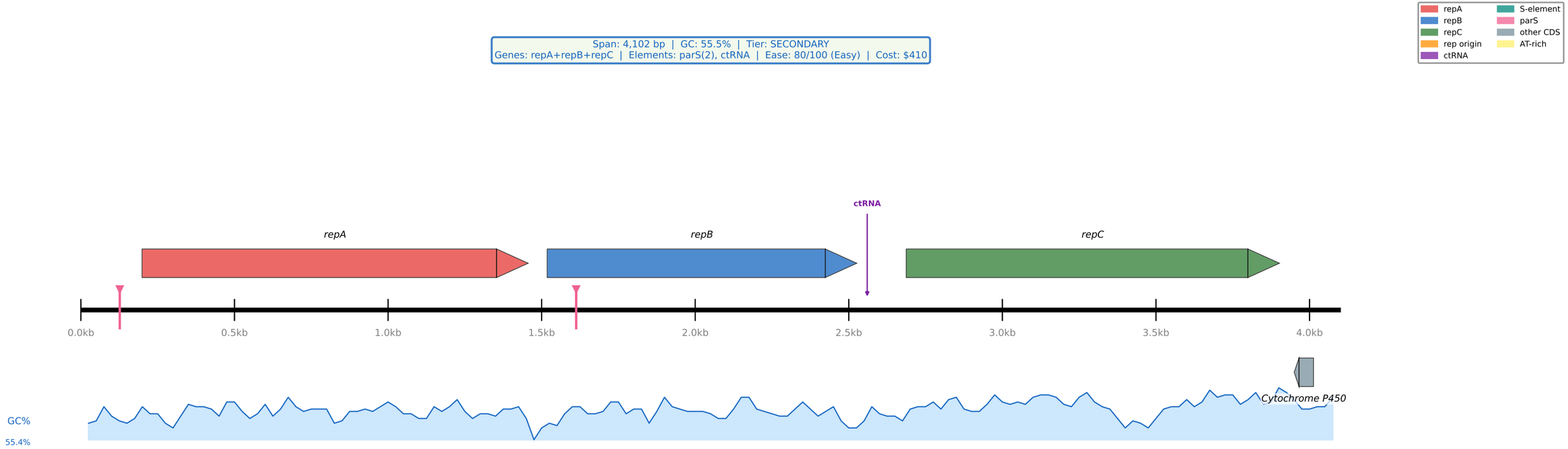
AF3.44_contig_4 | unknown | Agrobacterium tumefaciens

Span: 4,067 bp | GC: 55.6% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(1), ctRNA | Ease: 80/100 (Easy) | Cost: $407



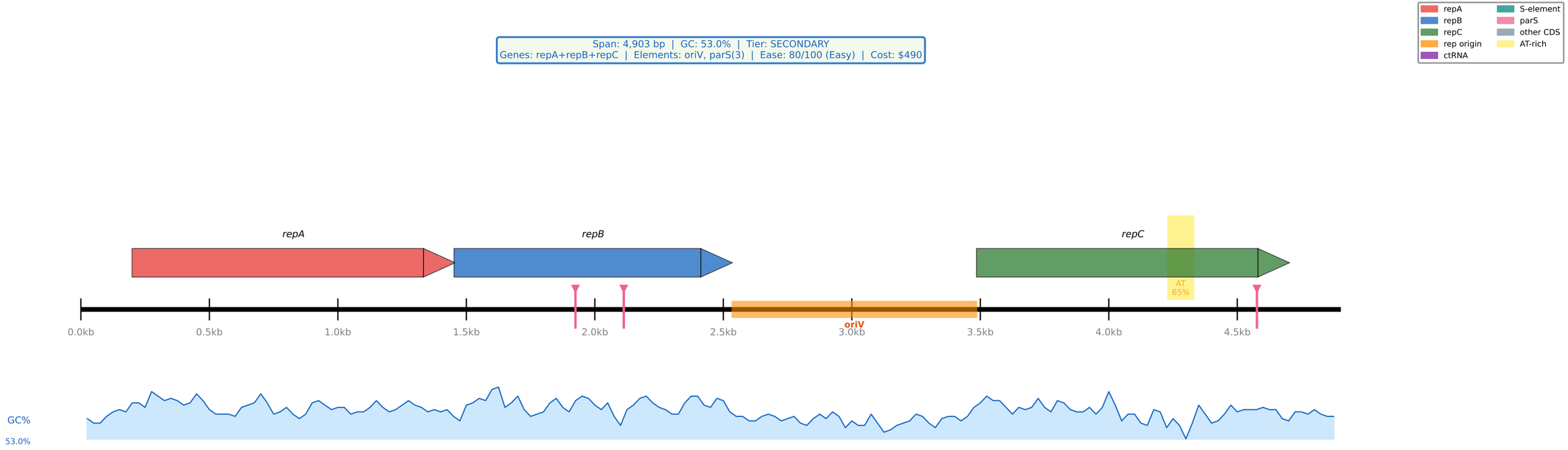
NCPPB_1641_NZ_LT009778.1 | unknown | Agrobacterium vitis

Span: 4,102 bp | GC: 55.5% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(2), ctRNA | Ease: 80/100 (Easy) | Cost: $410



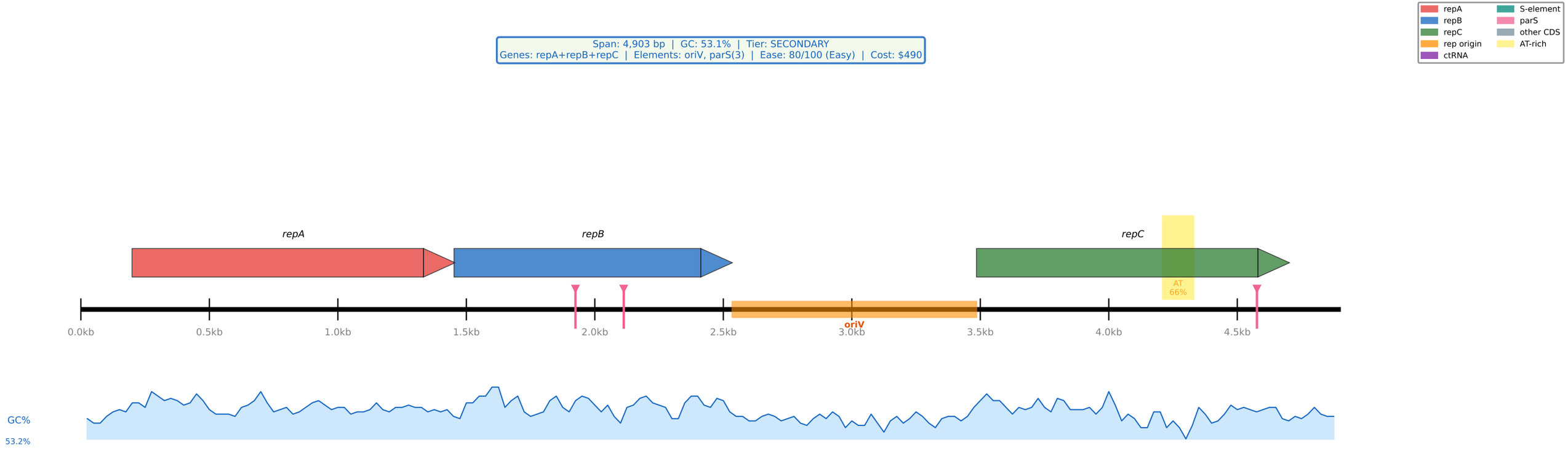
CG474_chromid | chromid | Agrobacterium tumefaciens

Span: 4,903 bp | GC: 53.0% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: oriV, parS(3) | Ease: 80/100 (Easy) | Cost: $490



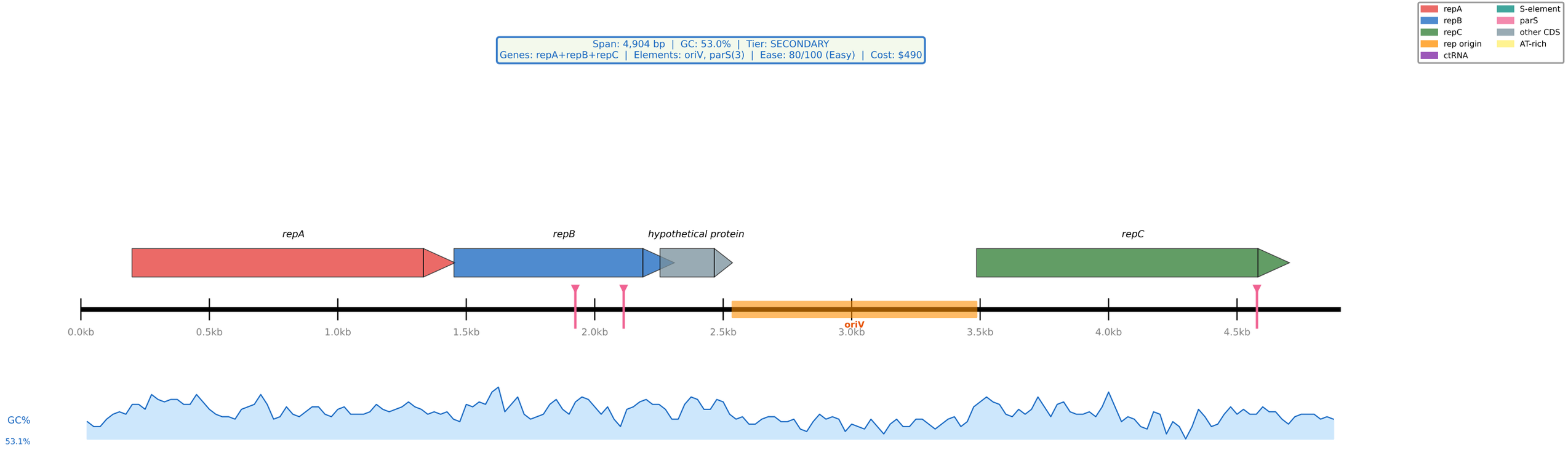
1D1514_chromid | chromid | Agrobacterium tumefaciens

Span: 4,903 bp | GC: 53.1% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: oriV, parS(3) | Ease: 80/100 (Easy) | Cost: $490



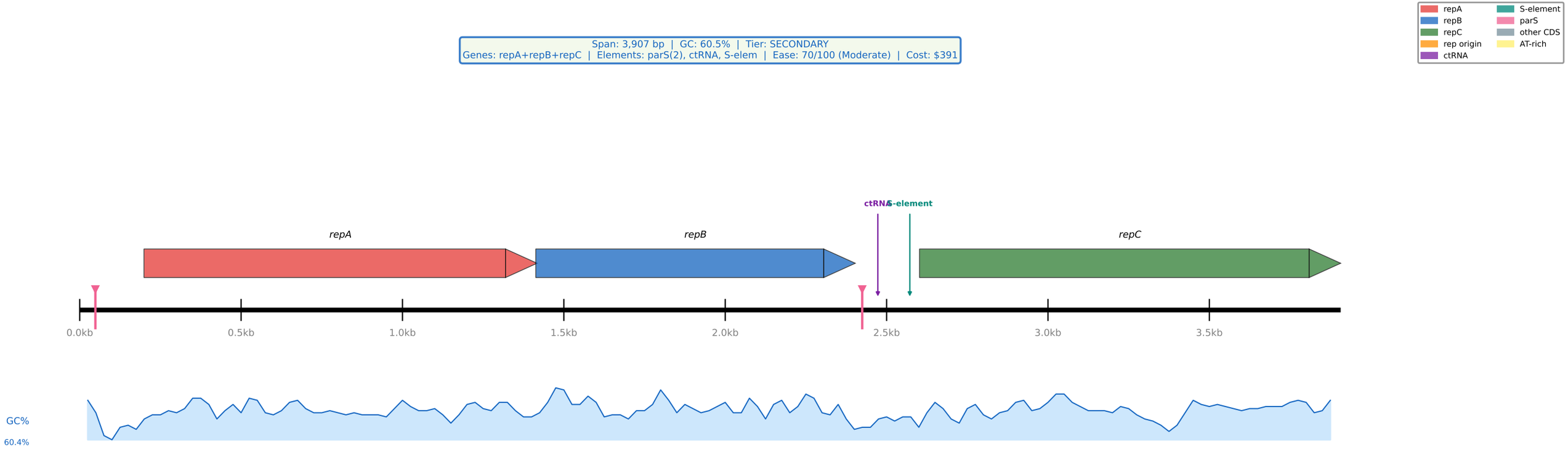
AMS008_chromid | chromid | Agrobacterium tumefaciens

Span: 4,904 bp | GC: 53.0% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: oriV, parS(3) | Ease: 80/100 (Easy) | Cost: $490



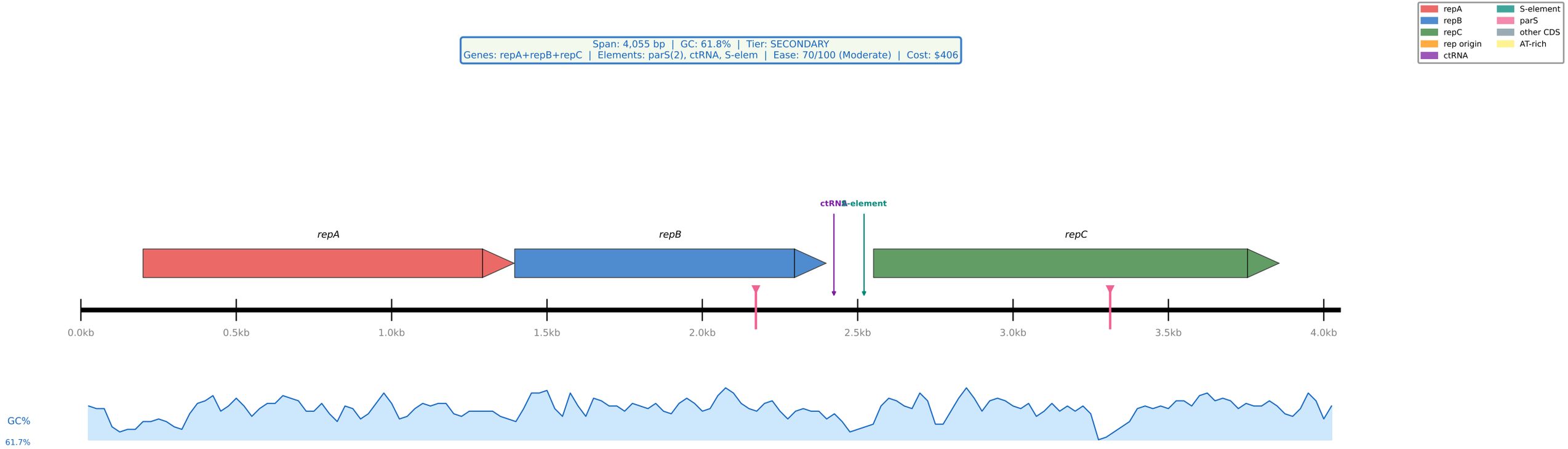
Rhizobium_etli_CFN_42_CP000136.1 | unknown | Rhizobium etli CFN 42

Span: 3,907 bp | GC: 60.5% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 70/100 (Moderate) | Cost: $391



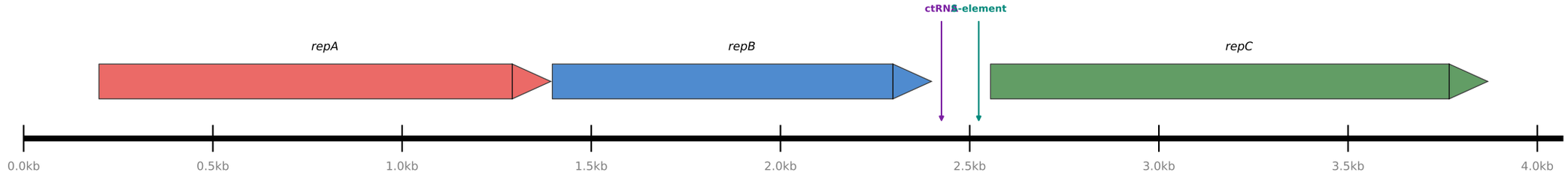
Rhizobium_etli_CFN_42_CP000138.1_rep_1 | unknown | Rhizobium etli CFN 42

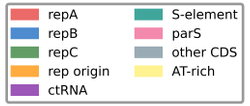
Span: 4,055 bp | GC: 61.8% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 70/100 (Moderate) | Cost: $406



NCIB_8196_NZ_JAAMDI010000042.1 | unknown | Agrobacterium rhizogenes

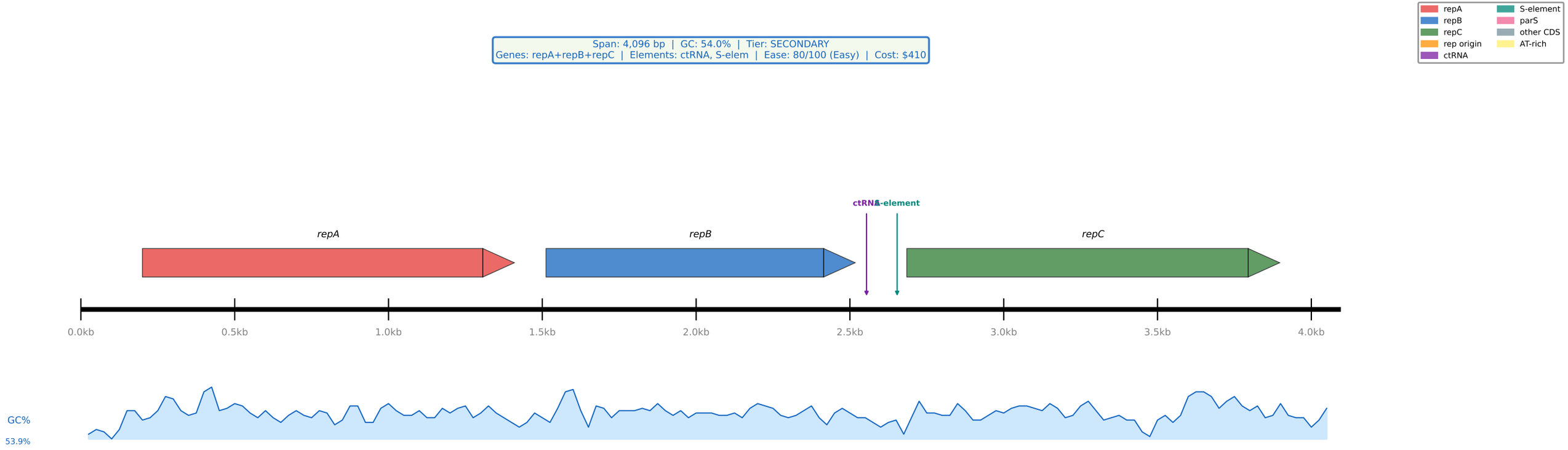
Span: 4,069 bp | GC: 54.8% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $407





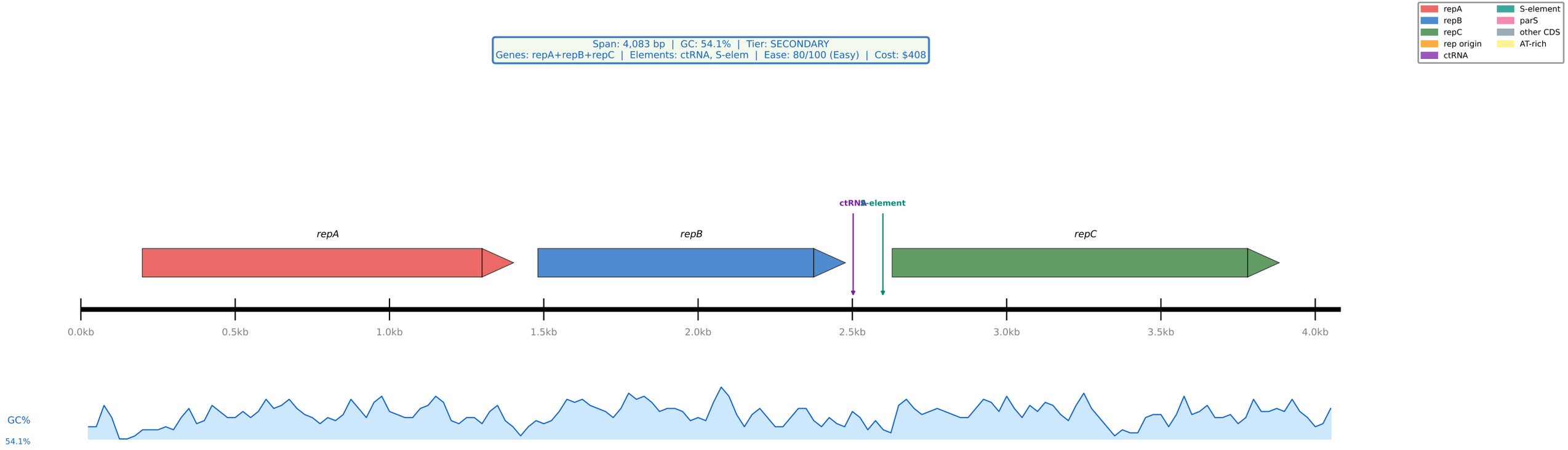
B21_90_contig_5 | unknown | Agrobacterium tumefaciens

Span: 4,096 bp | GC: 54.0% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $410



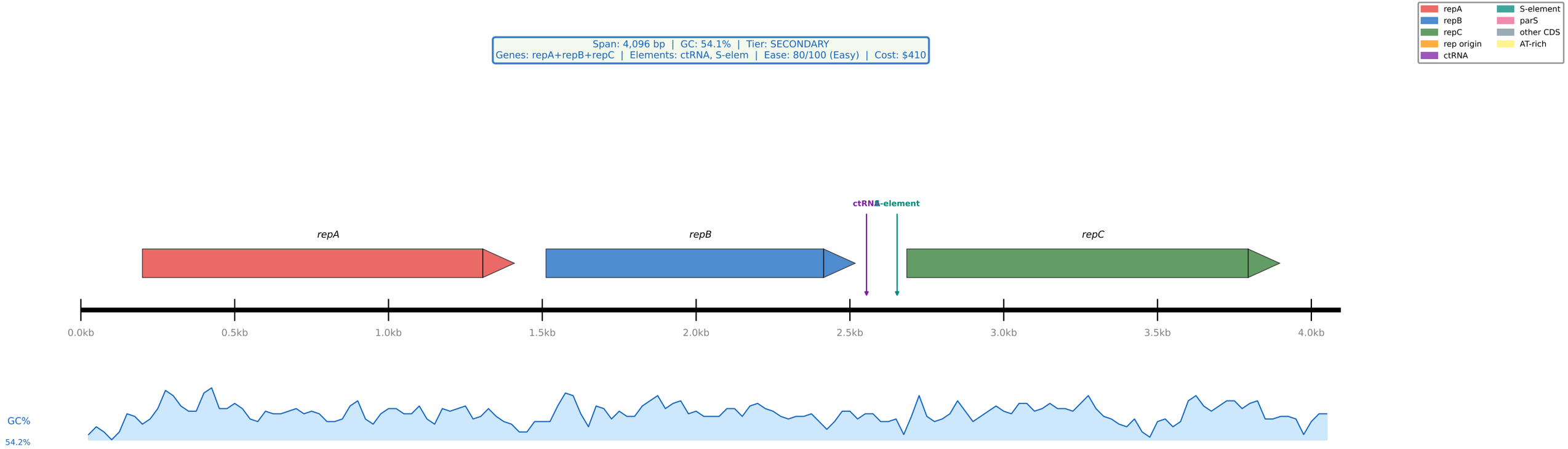
AB2_73_contig_4_pTi_Type_IX | Type IX | Agrobacterium tumefaciens

Span: 4,083 bp | GC: 54.1% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $408



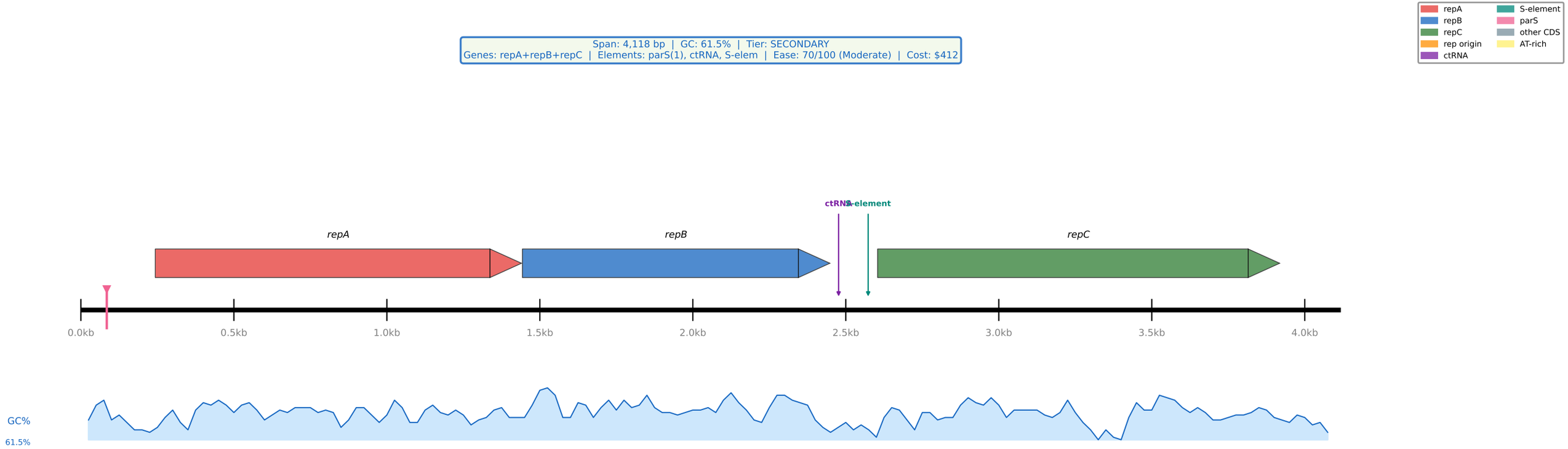
CG958_CG958_5 | unknown | Agrobacterium tumefaciens

Span: 4,096 bp | GC: 54.1% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $410



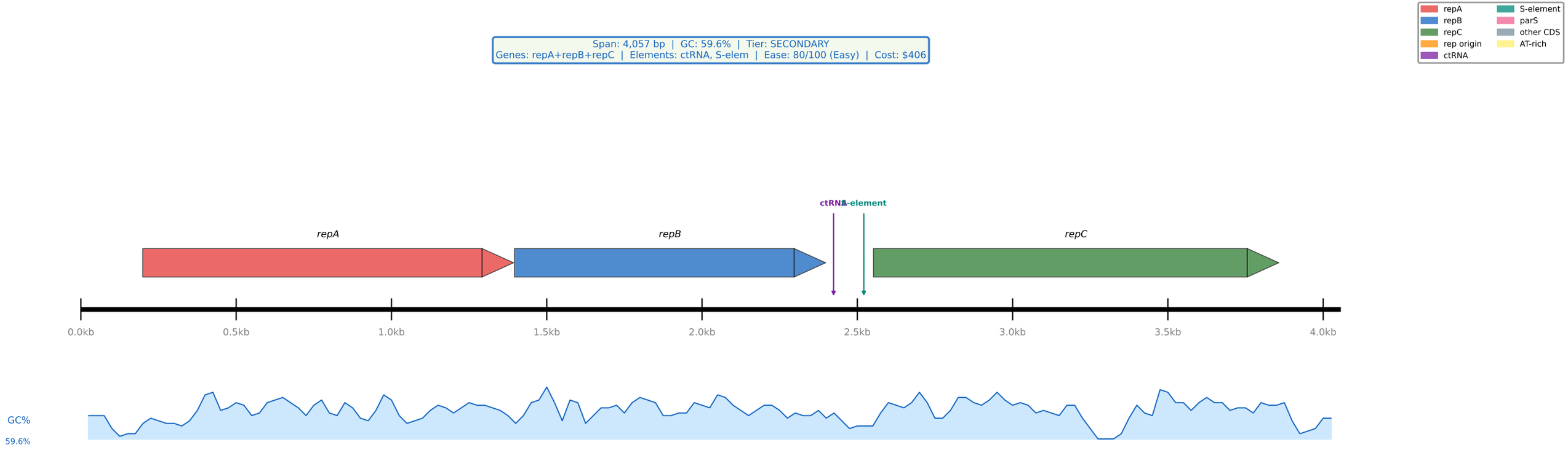
Sinorhizobium_fredii_NGR234_CP000874.1 | unknown | Sinorhizobium fredii NGR234

Span: 4,118 bp | GC: 61.5% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(1), ctRNA, S-elem | Ease: 70/100 (Moderate) | Cost: $412



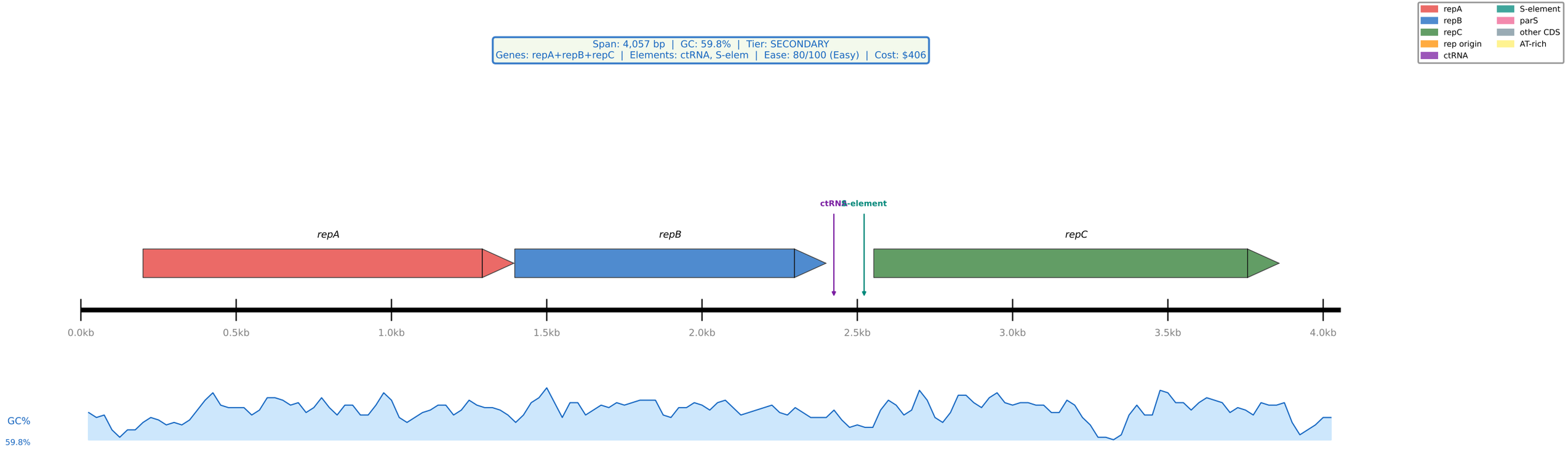
WS16_contig_3 | unknown | Agrobacterium tumefaciens

Span: 4,057 bp | GC: 59.6% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $406



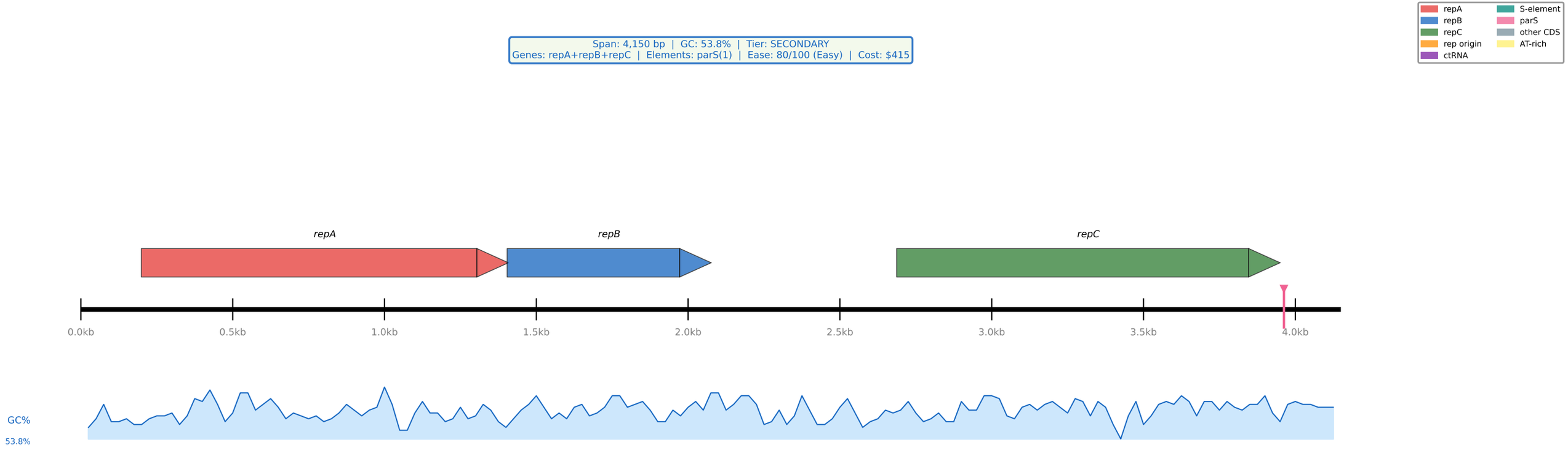
WS192_contig_5 | unknown | Agrobacterium tumefaciens

Span: 4,057 bp | GC: 59.8% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $406



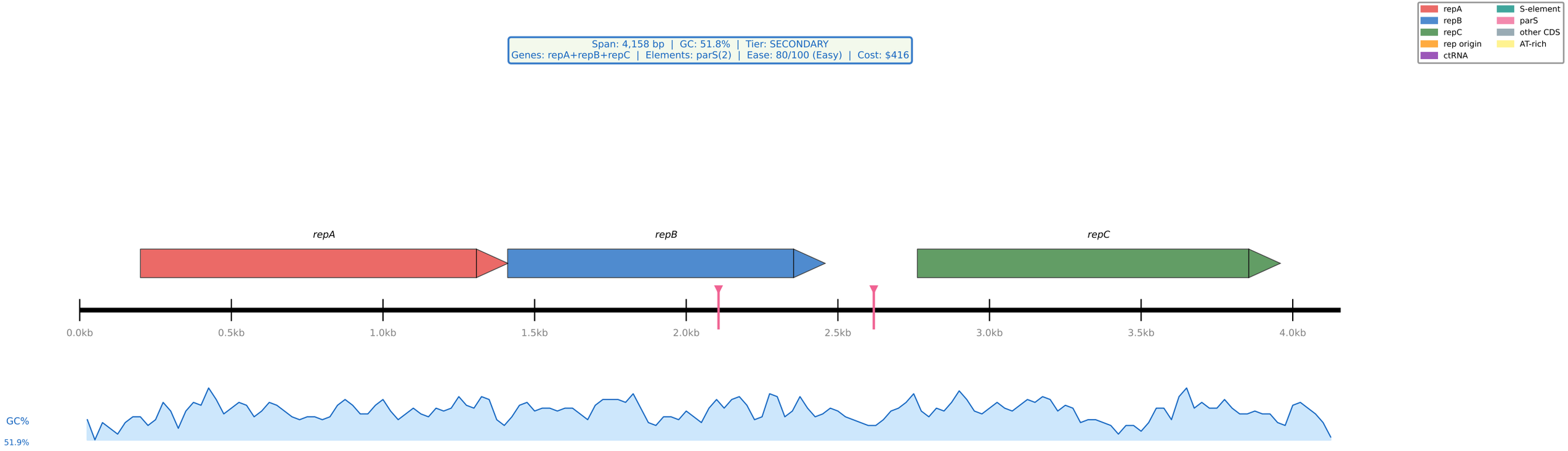
Rhizobium_etli_CFN_42_CP000134.1_rep_1 | unknown | Rhizobium etli CFN 42

Span: 4,150 bp | GC: 53.8% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(1) | Ease: 80/100 (Easy) | Cost: $415



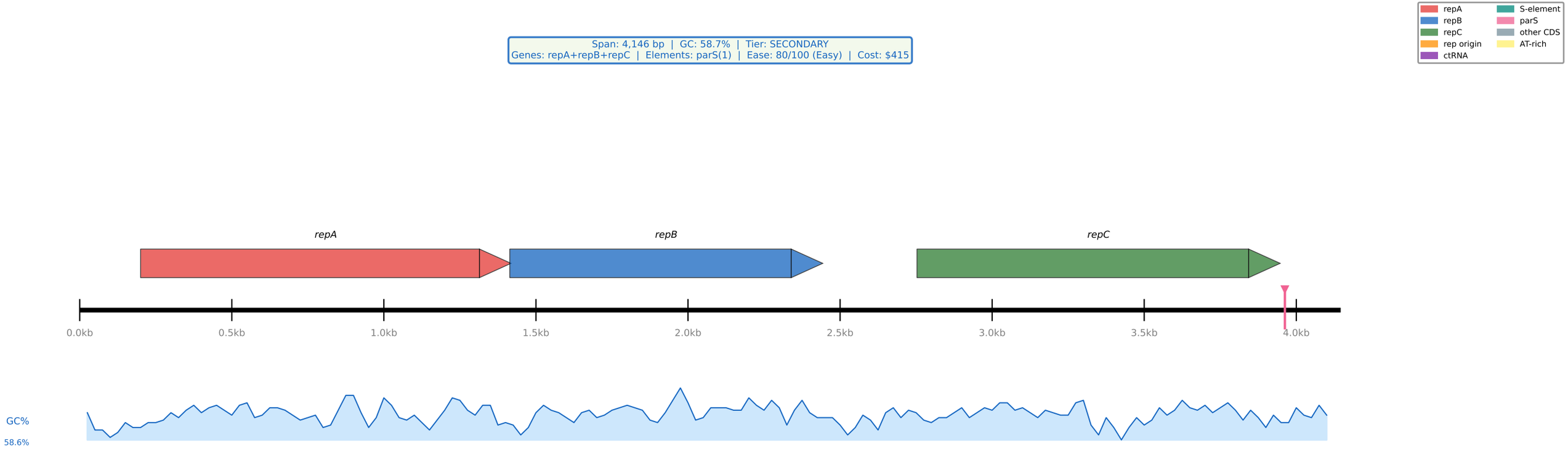
CG507_CG507_4 | unknown | Agrobacterium tumefaciens

Span: 4,158 bp | GC: 51.8% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(2) | Ease: 80/100 (Easy) | Cost: $416



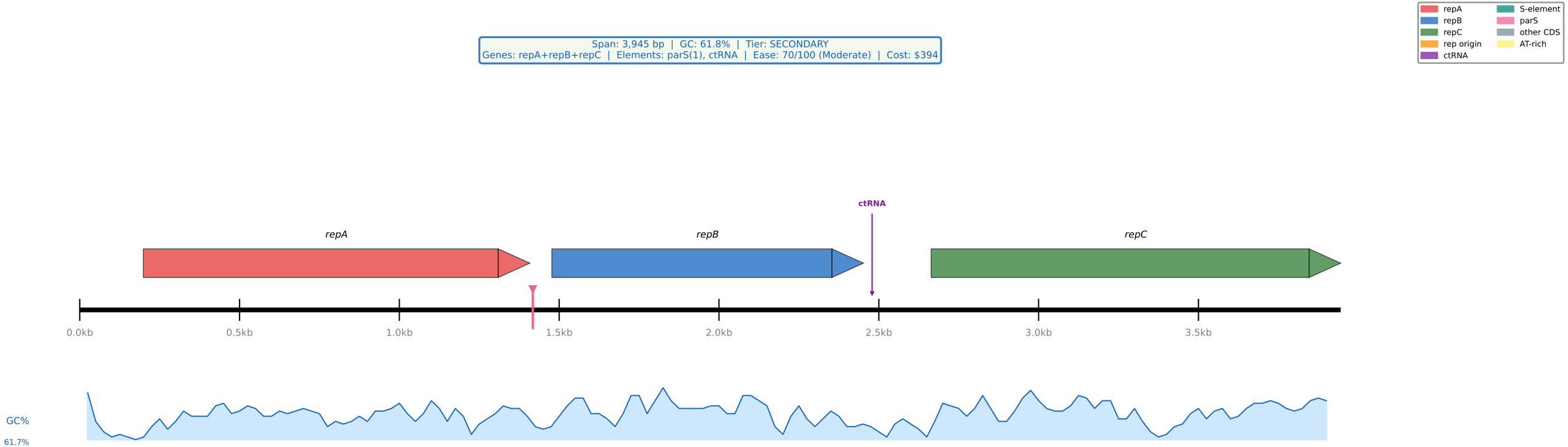
WS192_contig_3 | unknown | Agrobacterium tumefaciens

Span: 4,146 bp | GC: 58.7% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(1) | Ease: 80/100 (Easy) | Cost: $415



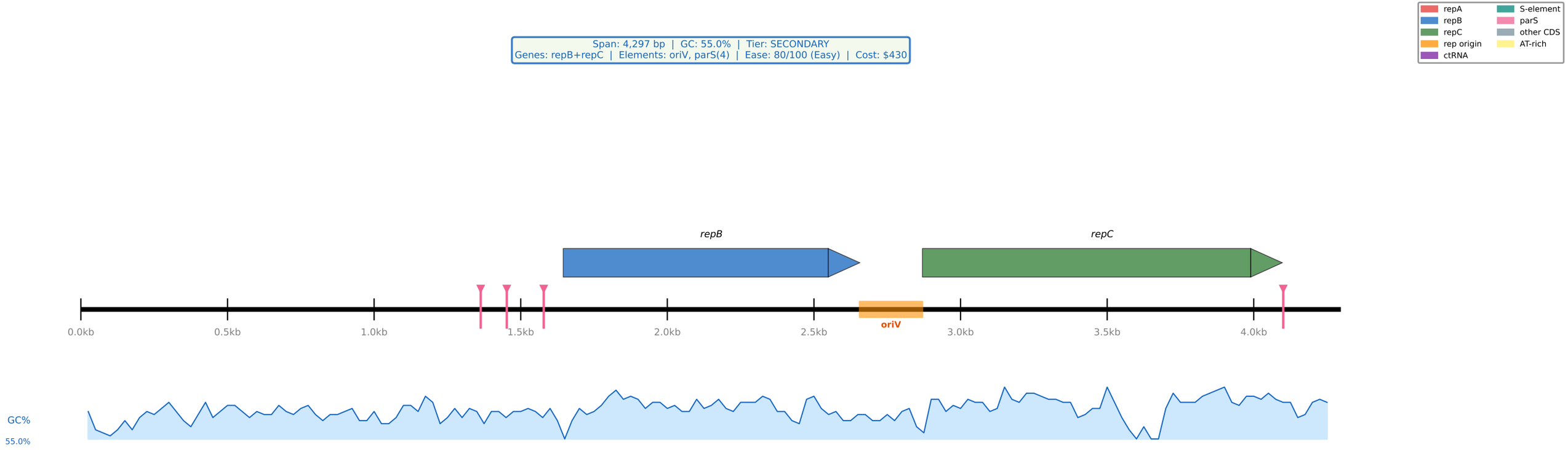
Rhizobium_etli_CFN_42_CP000138.1_rep_2 | unknown | Rhizobium etli CFN 42

Span: 3,945 bp | GC: 61.8% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(1), ctRNA | Ease: 70/100 (Moderate) | Cost: $394



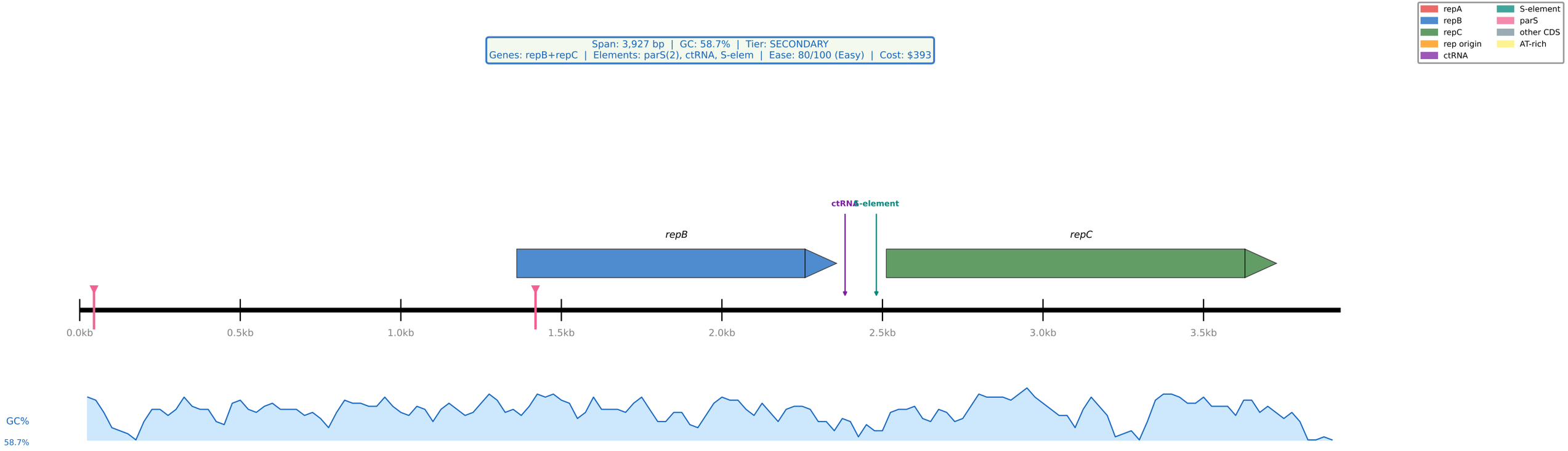
CG474_CG474_4_rep_1 | unknown | Agrobacterium tumefaciens

Span: 4,297 bp | GC: 55.0% | Tier: SECONDARY
Genes: repB+repC | Elements: oriV, parS(4) | Ease: 80/100 (Easy) | Cost: $430



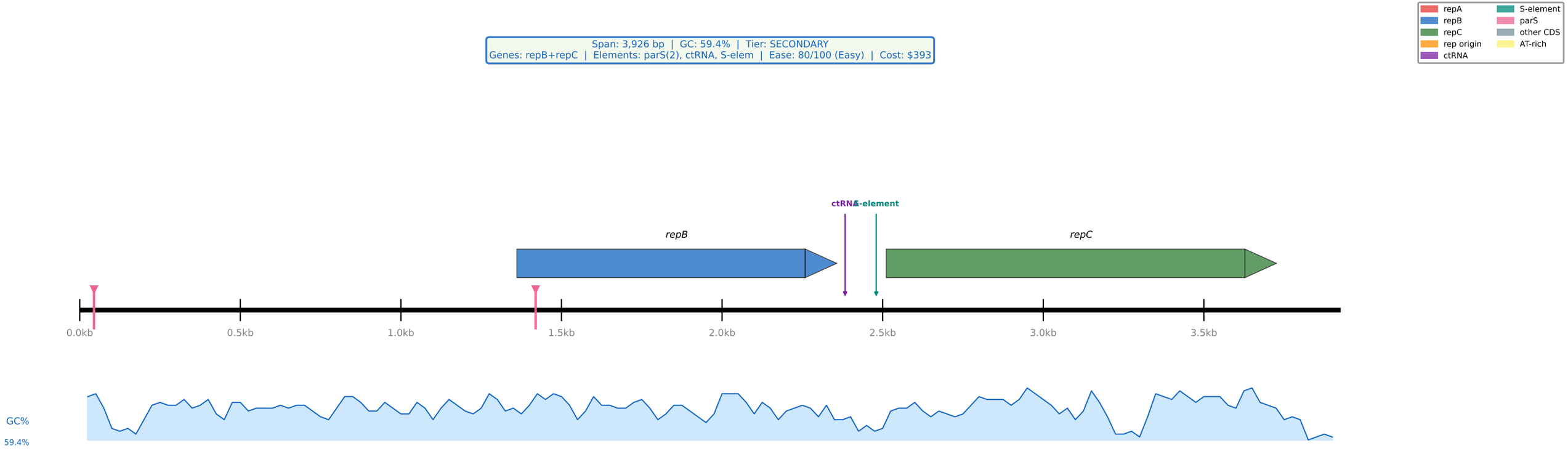
WS163_contig_4 | unknown | Agrobacterium tumefaciens

Span: 3,927 bp | GC: 58.7% | Tier: SECONDARY
Genes: repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $393



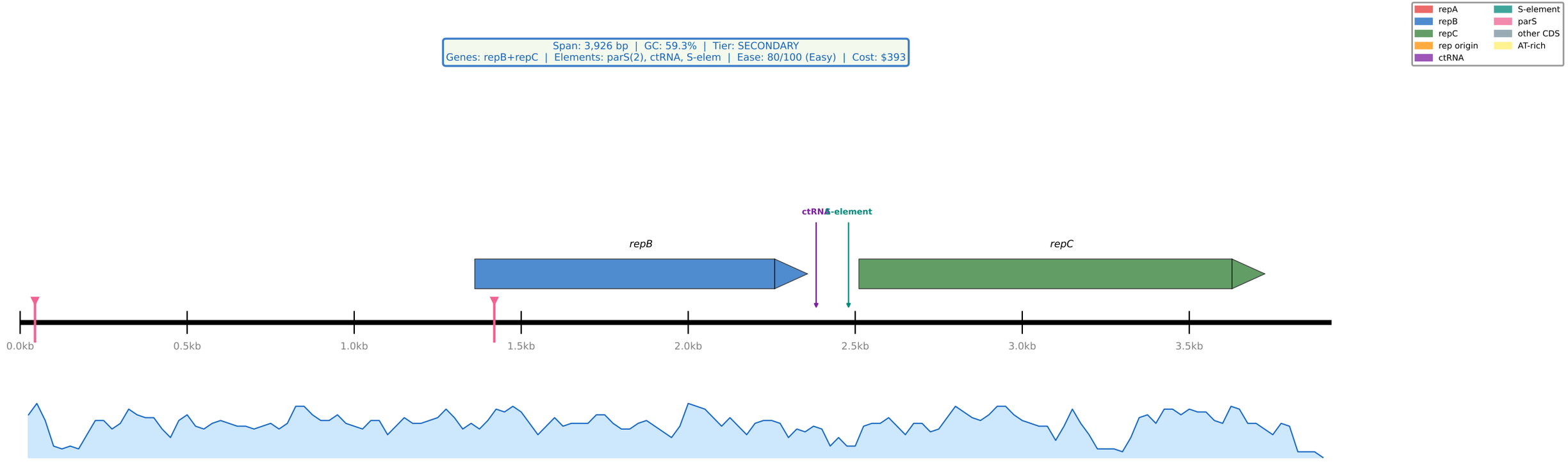
WS287_contig_3 | unknown | Agrobacterium tumefaciens

Span: 3,926 bp | GC: 59.4% | Tier: SECONDARY
Genes: repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $393



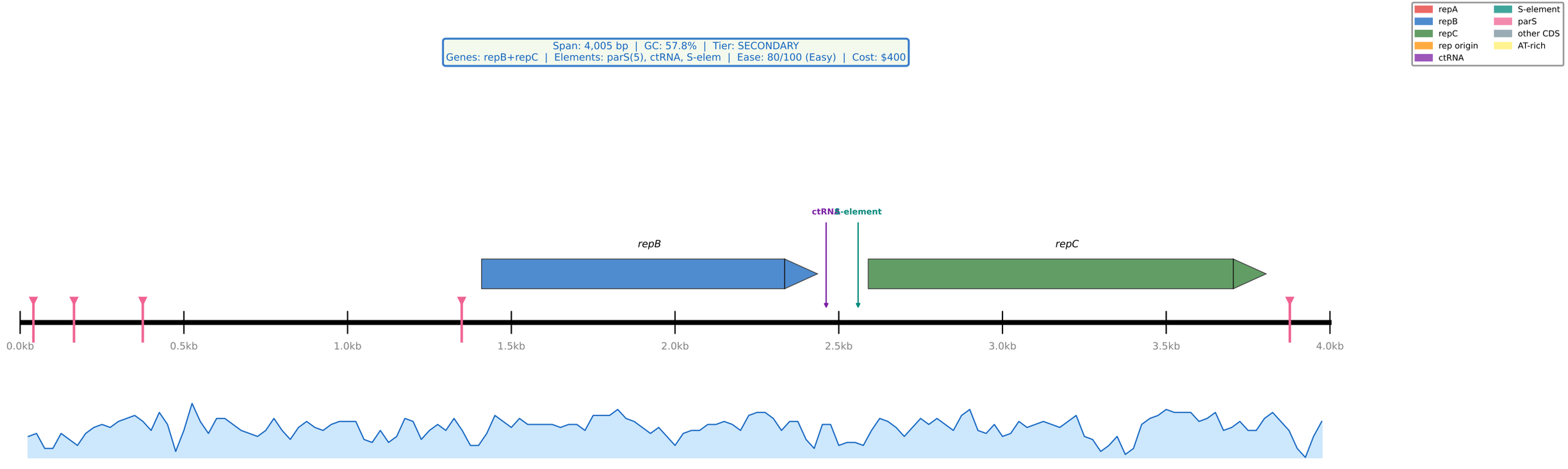
WS198_contig_3 | unknown | Agrobacterium tumefaciens

Span: 3,926 bp | GC: 59.3% | Tier: SECONDARY
Genes: repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $393



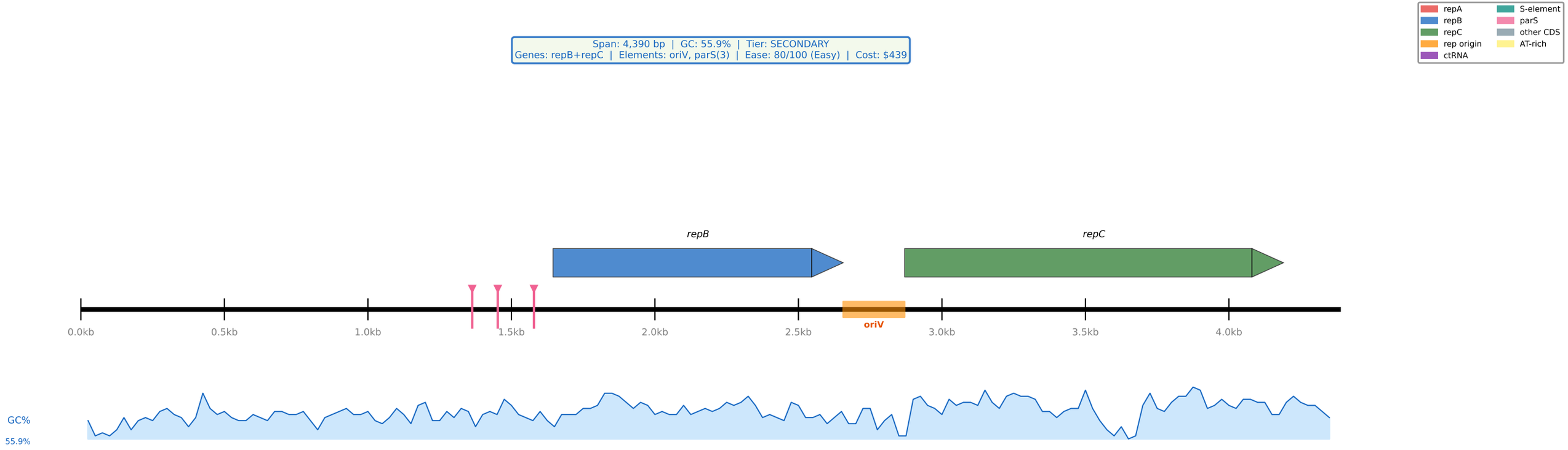
WS287_contig_6_rep_1 | unknown | Agrobacterium tumefaciens

Span: 4,005 bp | GC: 57.8% | Tier: SECONDARY
Genes: repB+repC | Elements: parS(5), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $400



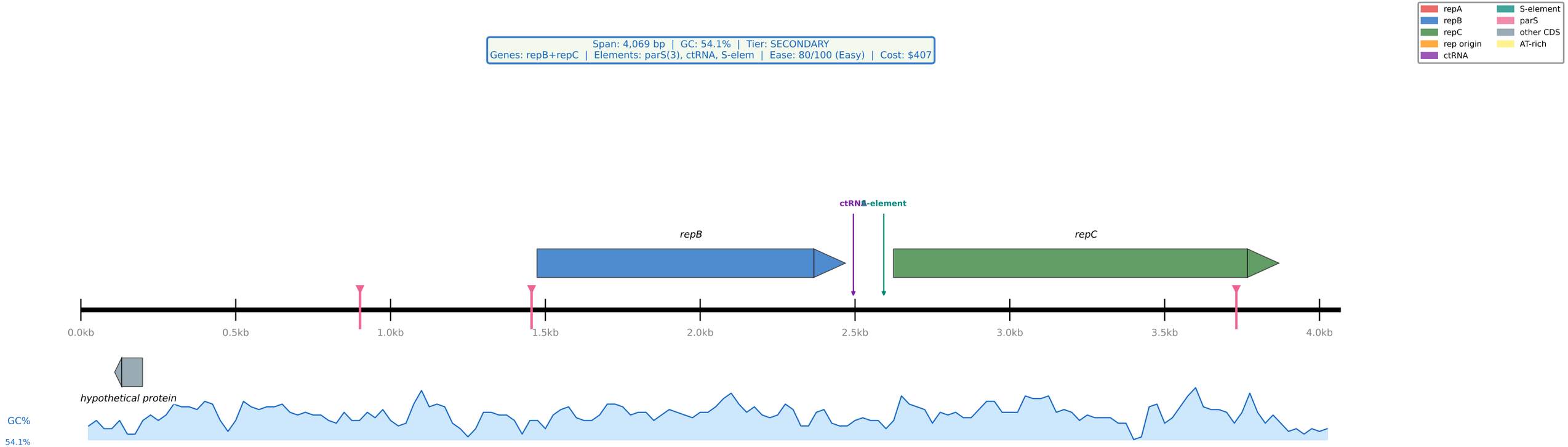
CG958_CG958_4_pTi_Type_IV.b | Type IV.b | Agrobacterium tumefaciens

Span: 4,390 bp | GC: 55.9% | Tier: SECONDARY
Genes: repB+repC | Elements: oriV, parS(3) | Ease: 80/100 (Easy) | Cost: $439



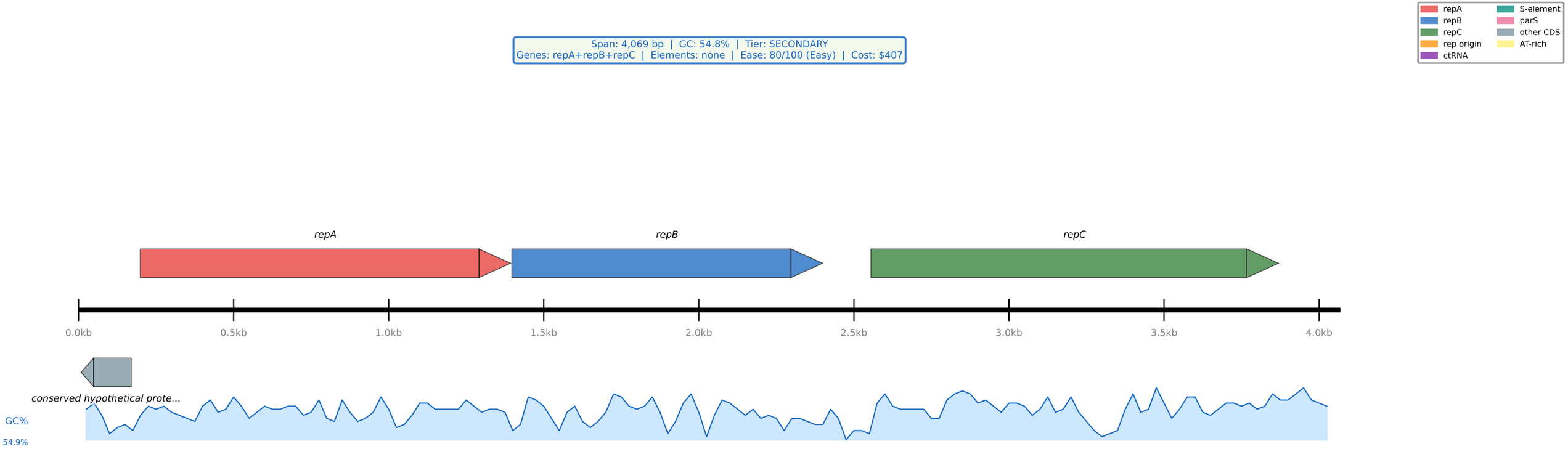
15-1187-1-2a_CP123841.1 | unknown | Agrobacterium tumefaciens

Span: 4,069 bp | GC: 54.1% | Tier: SECONDARY
Genes: repB+repC | Elements: parS(3), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $407



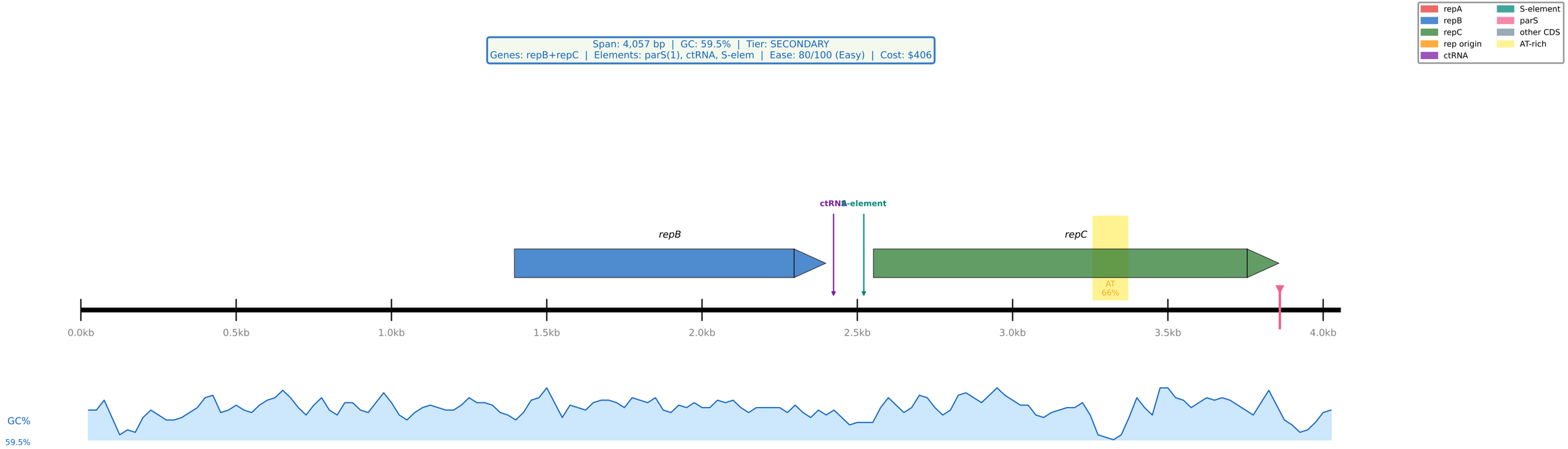
Atu_K84_pAtK84c_CP000631.1 | pAt | Agrobacterium radiobacter

Span: 4,069 bp | GC: 54.8% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: none | Ease: 80/100 (Easy) | Cost: $407



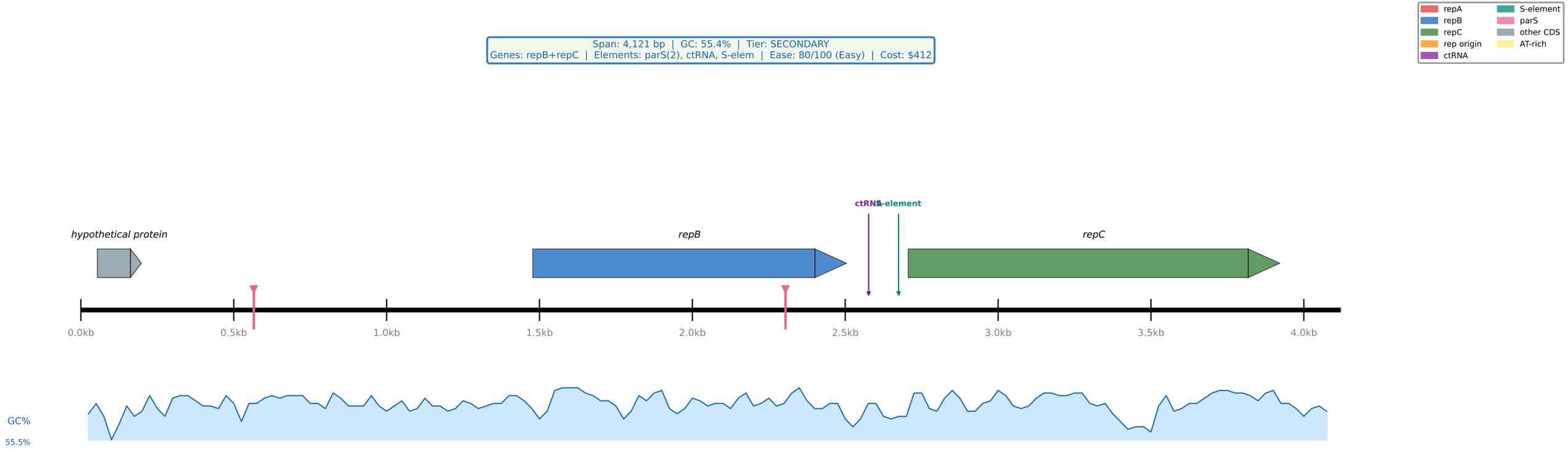
WS163_contig_3 | unknown | Agrobacterium tumefaciens

Span: 4,057 bp | GC: 59.5% | Tier: SECONDARY
Genes: repB+repC | Elements: parS(1), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $406



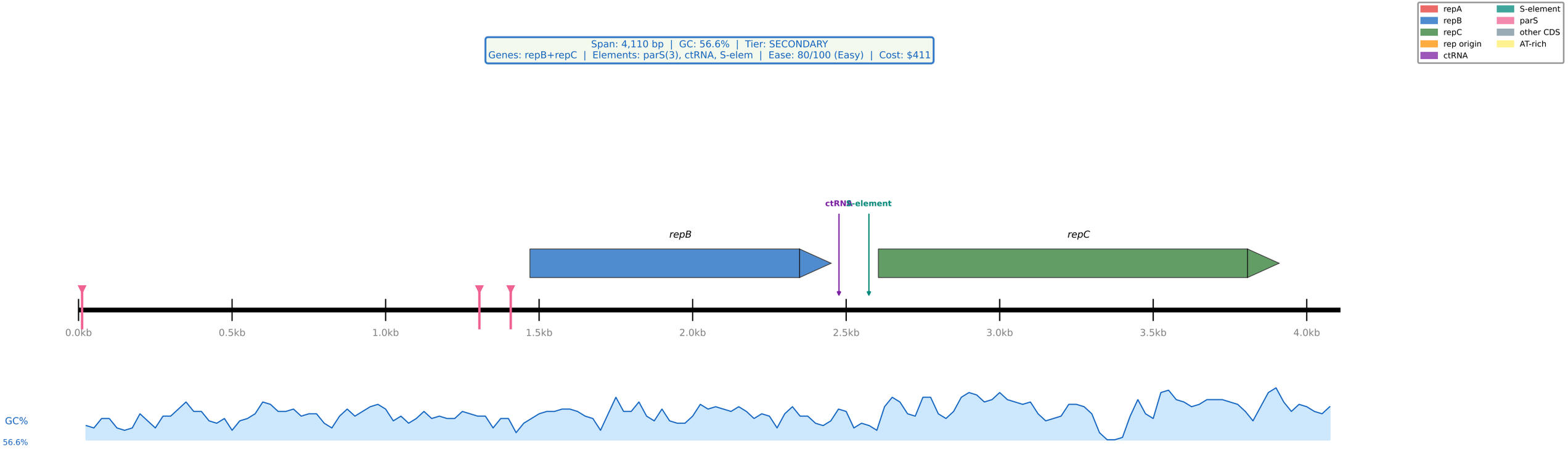
15-1187-1-2a_CP123840.1 | unknown | Agrobacterium tumefaciens

Span: 4,121 bp | GC: 55.4% | Tier: SECONDARY
Genes: repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $412



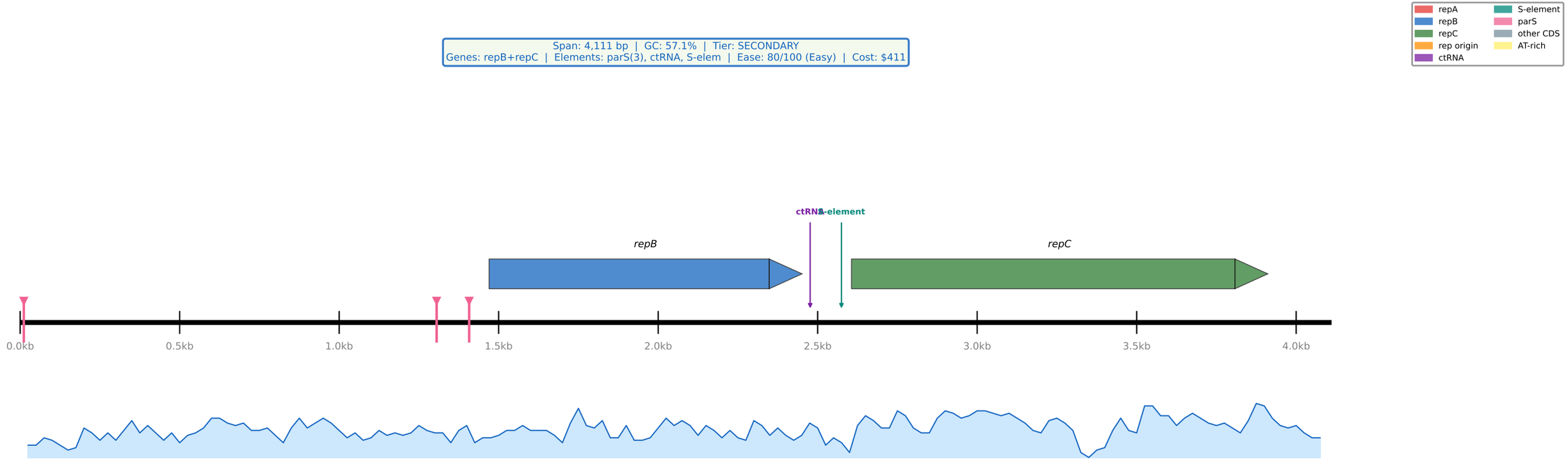
WS163_contig_6 | unknown | Agrobacterium tumefaciens

Span: 4,110 bp | GC: 56.6% | Tier: SECONDARY
Genes: repB+repC | Elements: parS(3), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $411



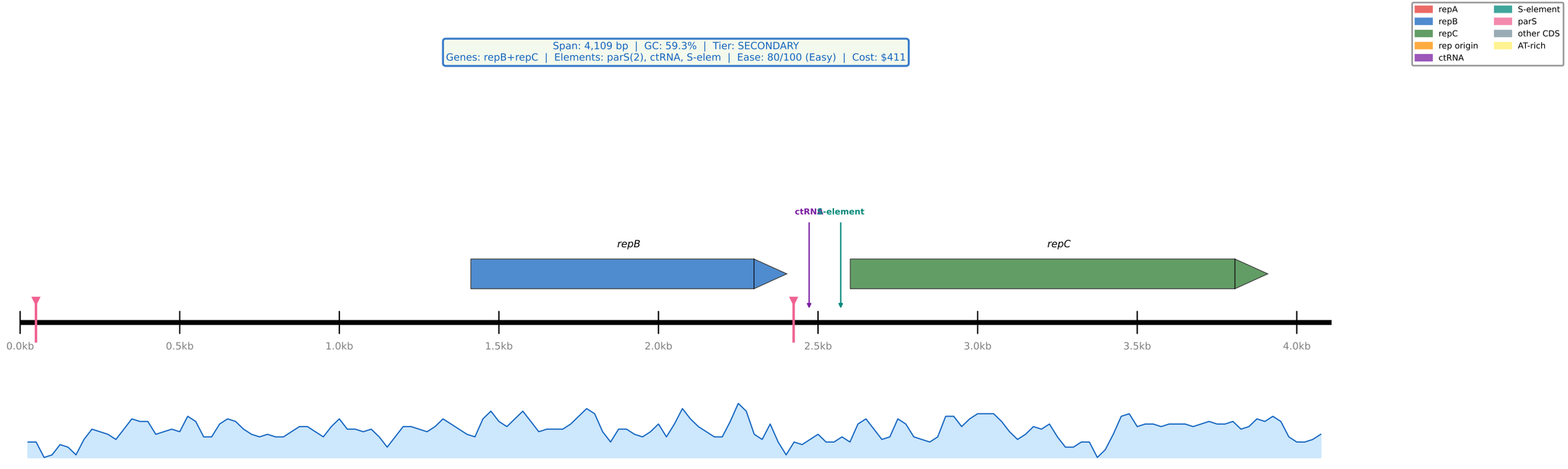
WS287_contig_5 | unknown | Agrobacterium tumefaciens

Span: 4,111 bp | GC: 57.1% | Tier: SECONDARY
Genes: repB+repC | Elements: parS(3), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $411



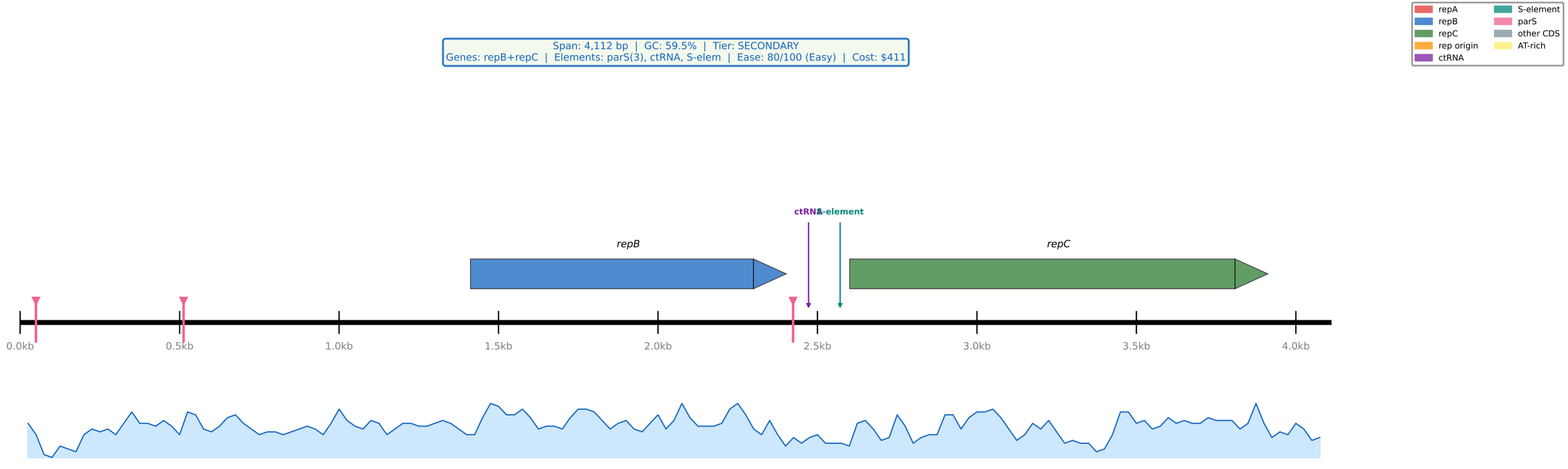
WS198_contig_4 | unknown | Agrobacterium tumefaciens

Span: 4,109 bp | GC: 59.3% | Tier: SECONDARY
Genes: repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $411



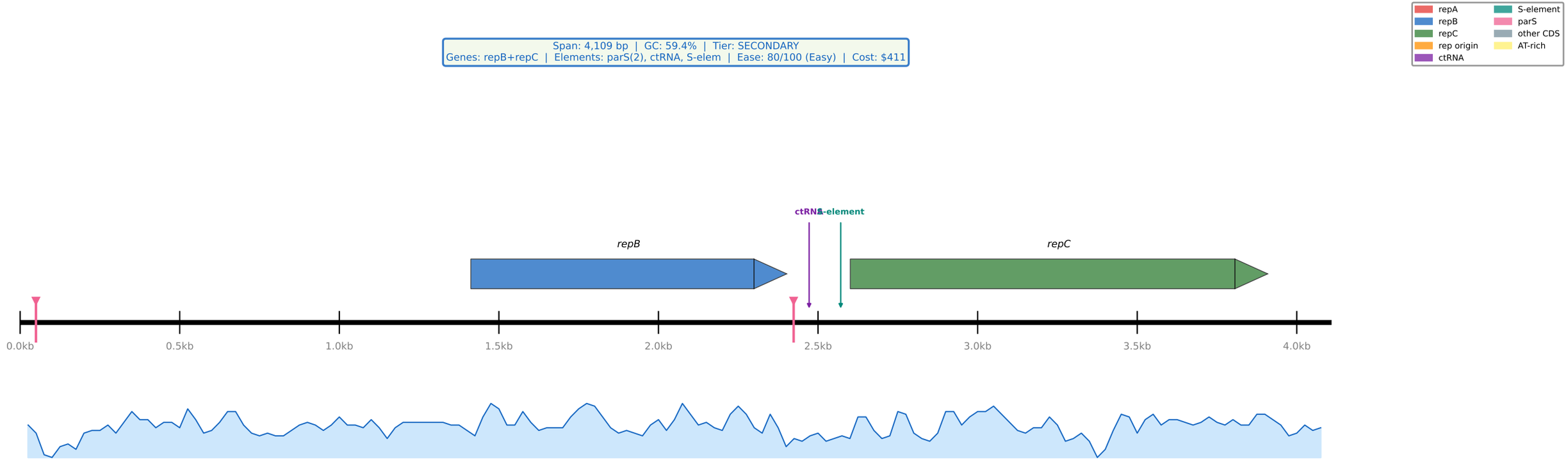
WS163_contig_5 | unknown | Agrobacterium tumefaciens

Span: 4,112 bp | GC: 59.5% | Tier: SECONDARY
Genes: repB+repC | Elements: parS(3), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $411



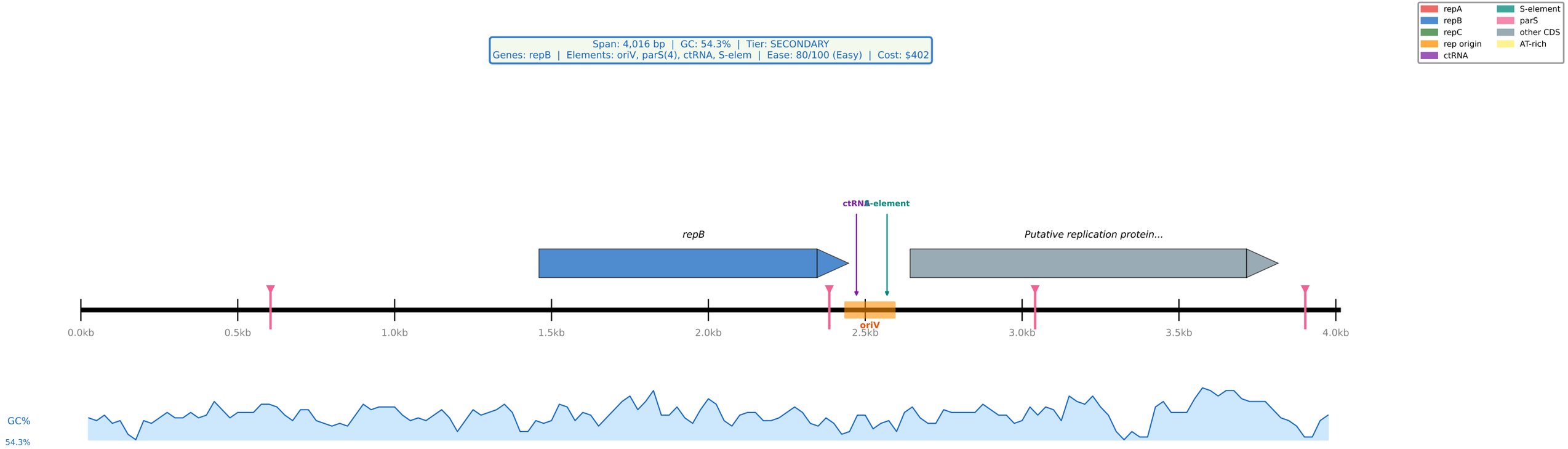
WS287_contig_4 | unknown | Agrobacterium tumefaciens

Span: 4,109 bp | GC: 59.4% | Tier: SECONDARY
Genes: repB+repC | Elements: parS(2), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $411



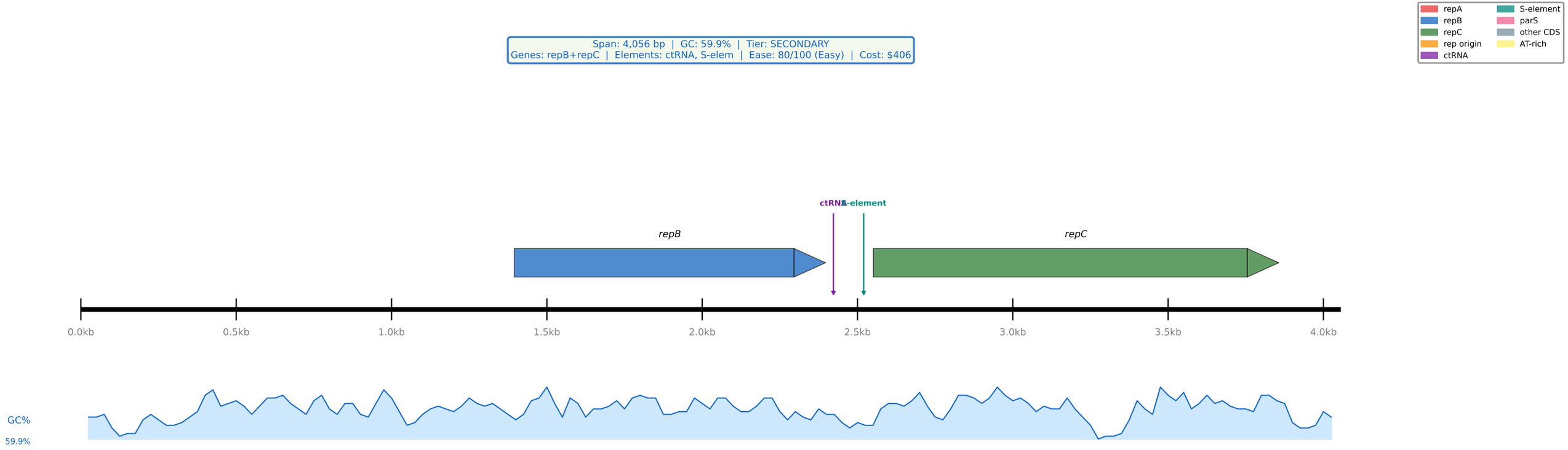
CG1055_contig_3 | unknown | Agrobacterium tumefaciens

Span: 4,016 bp | GC: 54.3% | Tier: SECONDARY
Genes: repB | Elements: oriV, parS(4), ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $402



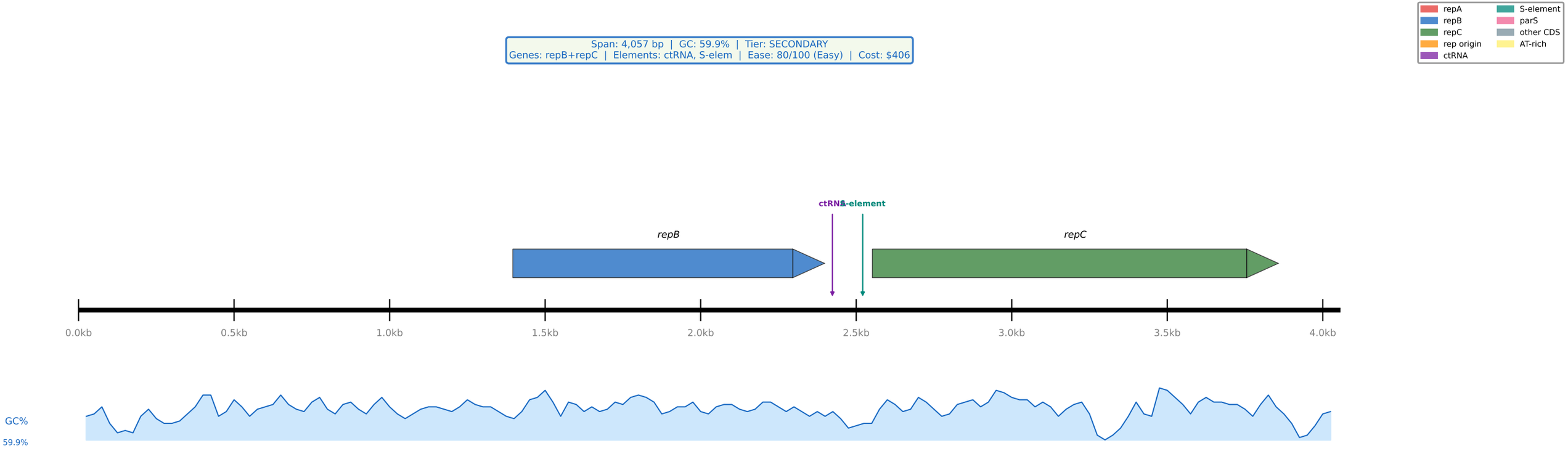
WS198_contig_2 | unknown | Agrobacterium tumefaciens

Span: 4,056 bp | GC: 59.9% | Tier: SECONDARY
Genes: repB+repC | Elements: ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $406



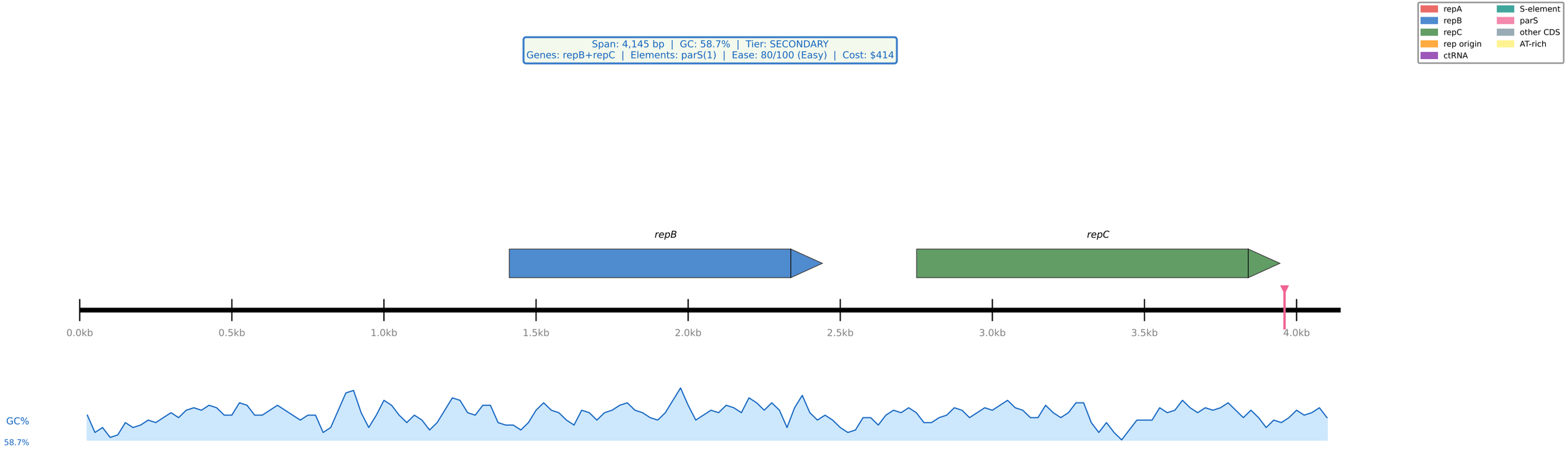
WS287_contig_2 | unknown | Agrobacterium tumefaciens

Span: 4,057 bp | GC: 59.9% | Tier: SECONDARY
Genes: repB+repC | Elements: ctRNA, S-elem | Ease: 80/100 (Easy) | Cost: $406



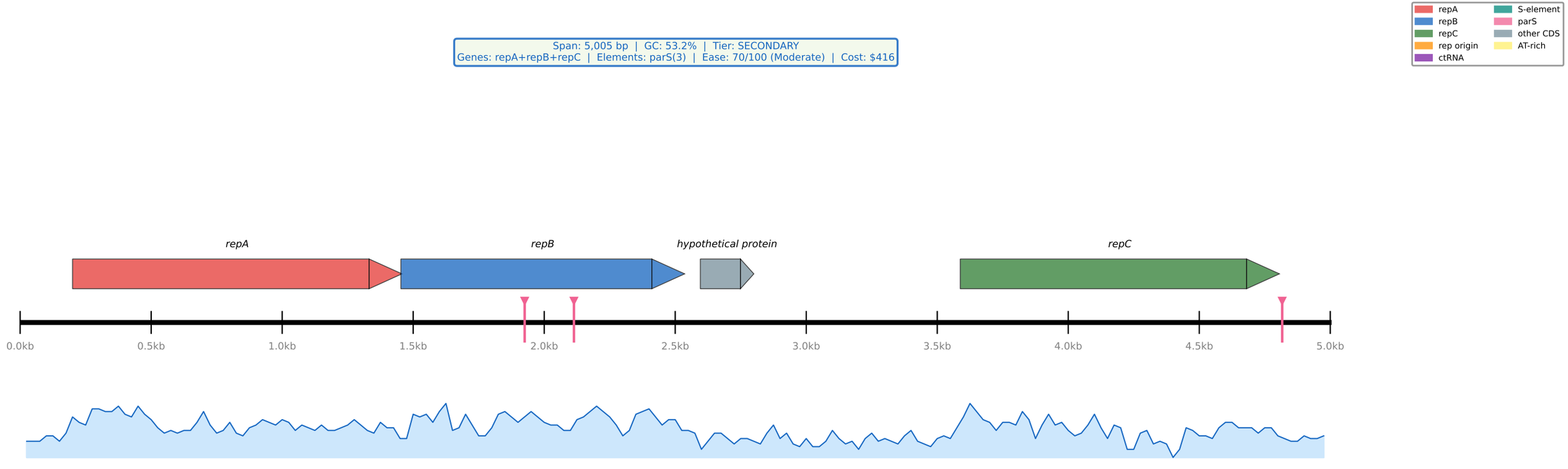
WS198_contig_5 | unknown | Agrobacterium tumefaciens

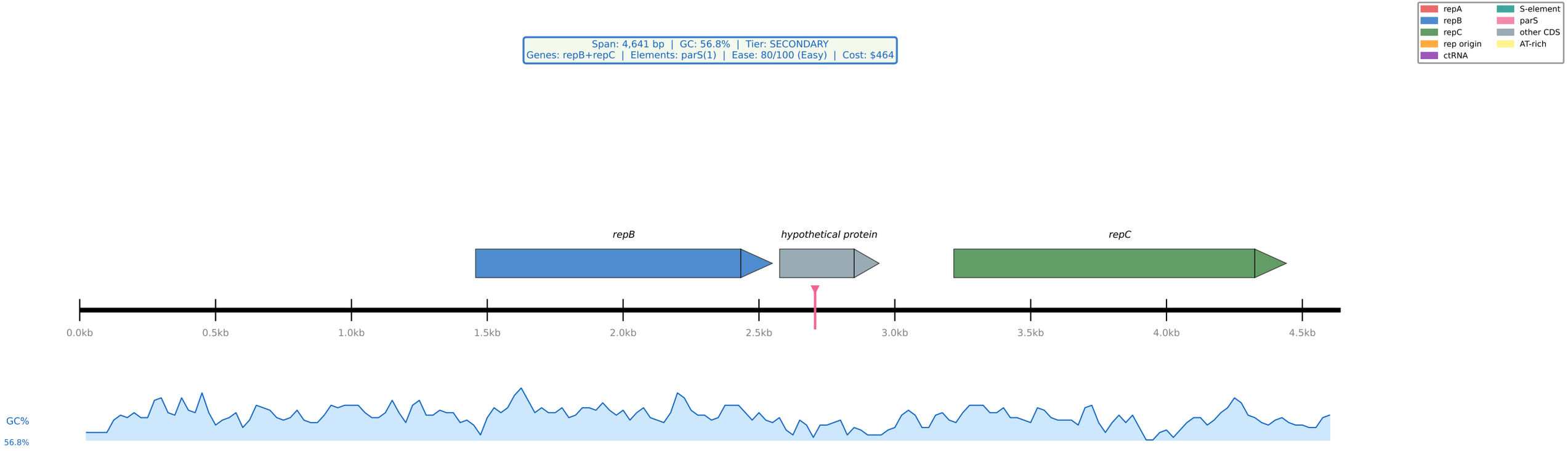
Span: 4,145 bp | GC: 58.7% | Tier: SECONDARY
Genes: repB+repC | Elements: parS(1) | Ease: 80/100 (Easy) | Cost: $414



CG958_chromid | chromid | Agrobacterium tumefaciens

Span: 5,005 bp | GC: 53.2% | Tier: SECONDARY
Genes: repA+repB+repC | Elements: parS(3) | Ease: 70/100 (Moderate) | Cost: $416



CG1055_contig_2 | unknown | Agrobacterium tumefaciens

Span: 4,641 bp | GC: 56.8% | Tier: SECONDARY
Genes: repB+repC | Elements: parS(1) | Ease: 80/100 (Easy) | Cost: $464